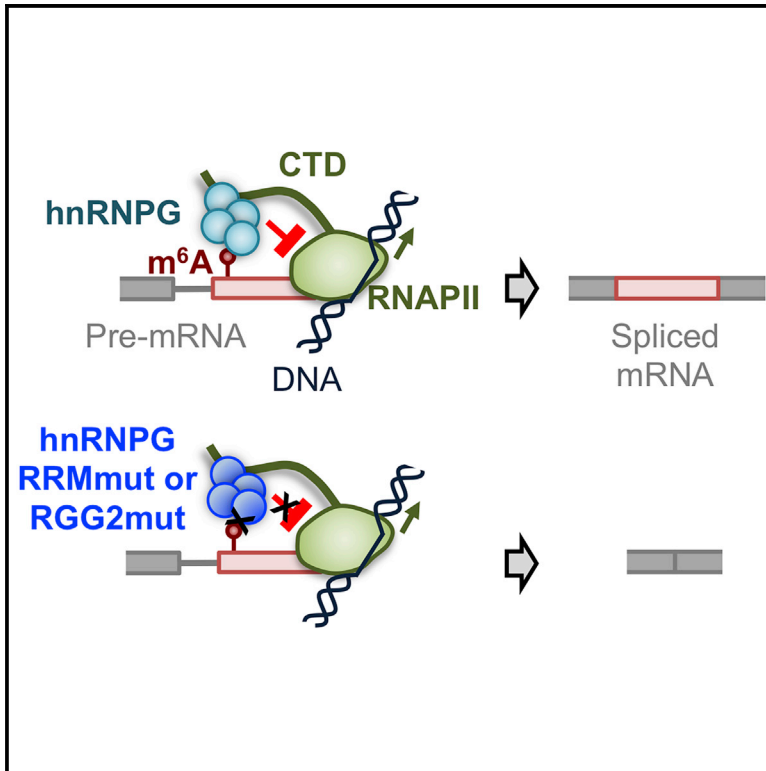


Regulation of Co-transcriptional Pre-mRNA Splicing by m⁶A through the Low-Complexity Protein hnRNPG

Graphical Abstract



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In Brief

Zhou et al. show that the m⁶A reader protein hnRNPG interacts with m⁶A-modified nascent pre-mRNA and the phosphorylated C-terminal domain of RNA polymerase II to regulate alternative splicing. These interactions depend on an RGG region in the low-complexity region of hnRNPG.

Highlights

- The m⁶A reader protein hnRNPG interacts with RNA polymerase II using an RGG region
- hnRNPG binds to nascent m⁶A-modified pre-mRNA and regulates alternative splicing
- m⁶A is enriched in exonic regions near splice sites of hnRNPG-regulated exons
- hnRNPG affects RNA polymerase II occupancy around hnRNPG-regulated exons



Regulation of Co-transcriptional Pre-mRNA Splicing by m⁶A through the Low-Complexity Protein hnRNPG

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<https://doi.org/10.1016/j.molcel.2019.07.005>

SUMMARY

N⁶-methyladenosine (m⁶A) modification occurs co-transcriptionally and impacts pre-mRNA processing; however, the mechanism of co-transcriptional m⁶A-dependent alternative splicing regulation is still poorly understood. Heterogeneous nuclear ribonucleoprotein G (hnRNPG) is an m⁶A reader protein that binds RNA through RRM and Arg-Gly-Gly (RGG) motifs. Here, we show that hnRNPG directly binds to the phosphorylated carboxy-terminal domain (CTD) of RNA polymerase II (RNAPII) using RGG motifs in its low-complexity region. Through interactions with the phosphorylated CTD and nascent RNA, hnRNPG associates co-transcriptionally with RNAPII and regulates alternative splicing transcriptome-wide. m⁶A near splice sites in nascent pre-mRNA modulates hnRNPG binding, which influences RNAPII occupancy patterns and promotes exon inclusion. Our results reveal an integrated mechanism of co-transcriptional m⁶A-mediated splicing regulation, in which an m⁶A reader protein uses RGG motifs to co-transcriptionally interact with both RNAPII and m⁶A-modified nascent pre-mRNA to modulate RNAPII occupancy and alternative splicing.

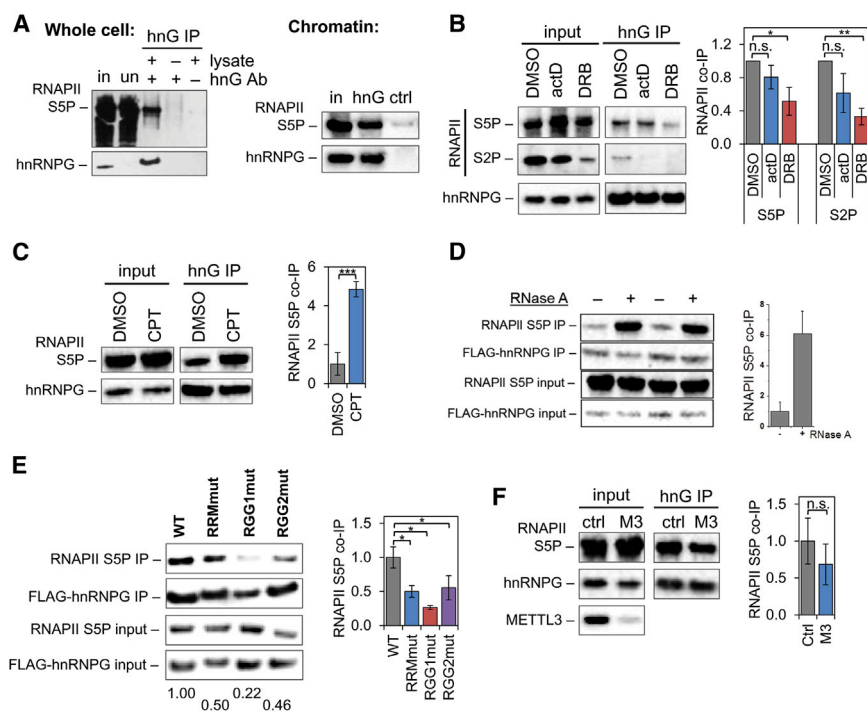
INTRODUCTION

The carboxy-terminal domain (CTD) of RNA polymerase II (RNAPII) plays a crucial role coordinating co-transcriptional processes with transcription (Bentley, 2014). The CTD is composed of tandem heptad repeats that undergo various post-translational modifications, including phosphorylation of Ser5 by cyclin-dependent kinase 7 (CDK7) during promoter escape and phosphorylation of

Ser2 by CDK9 upon promoter-proximal pause release (Harlen and Churchman, 2017). The various phosphorylation states of the CTD coordinate co-transcriptional processes with the stages of transcription: Ser5 phosphorylated CTD (S5P-CTD) recruits 5' capping factors at the 5' end of the gene; Ser2 phosphorylated CTD (S2P-CTD) recruits constitutive splicing factors in the gene body and 3' cleavage and polyadenylation factors at the 3' end of the gene (Hsin and Manley, 2012). While the CTD also contributes to alternative splicing, few alternative splicing factors are known to directly interact with the CTD (Kornblihtt et al., 2013; Muñoz et al., 2010). Although alternative splicing factors have been proposed to regulate splicing by modulating RNAPII pausing (Kornblihtt et al., 2013), previous studies have primarily examined this mode of splicing regulation at the transcriptome or interactome level (Ip et al., 2011; Shukla et al., 2011) and have not yet elucidated precise molecular mechanisms.

The most abundant and reversible internal modification in eukaryotic mRNA is N⁶-methyladenosine (m⁶A). m⁶A is deposited co-transcriptionally (Barbieri et al., 2017; Huang et al., 2019; Ke et al., 2017; Knuckles et al., 2017; Louloui et al., 2018) at a subset of RRACH (R = A/G, H = A/C/U) sequences (Dominissini et al., 2012; Meyer et al., 2012) by an m⁶A methyltransferase complex (writer) with core components methyltransferase-like 3 (METTL3) and METTL14 (Liu et al., 2014). The m⁶A modification of an mRNA transcript can influence every step in its life cycle, including alternative splicing (Xiao et al., 2016; Liu et al., 2015, 2017; Roignant and Soller, 2017): (1) the m⁶A writers METTL3 and METTL14 localize in part to nuclear speckles (Liu et al., 2014; Ping et al., 2014; Wang et al., 2014), and their depletion leads to changes in alternative splicing (Liu et al., 2014, 2015; Louloui et al., 2018; Ping et al., 2014; Yoon et al., 2017); (2) the m⁶A erasers FTO and ALKBH5 also localize in part to nuclear speckles (Jia et al., 2011; Zheng et al., 2013), and their depletion leads to changes in alternative splicing (Bartosovic et al., 2017; Qiao et al., 2016; Tang et al., 2018; Zhao et al., 2014); and (3) depletion of m⁶A readers YTHDC1, heterogeneous nuclear ribonucleoprotein C (hnRNPC), hnRNPG, and hnRNPA2B1 leads to changes in





hnRNPG knockdown (Figure S1A) and was reproducible with three different hnRNPG antibodies (Figure S1B). Upon transcription inhibition with actinomycin D (actD) or 5,6-dichloro-1- β -D-ribofuranosylbenzimidazole (DRB), co-immunoprecipitation of RNAPII with hnRNPG decreased (Figure 1B); upon inhibition with camptothecin (CPT), co-immunoprecipitation of RNAPII increased (Figure 1C). While all three inhibitors block transcription elongation, actinomycin D and DRB decrease, whereas camptothecin increases, occupancy of transcribing RNAPII on chromatin (Borsos et al., 2017; Ljungman and Hanawalt, 1996; Sordet et al., 2008). Thus, hnRNPG likely interacts with transcribing, chromatin-bound RNAPII. Consistent with this result, inhibiting RNAPII transcription with α -amanitin or actinomycin D altered the localization of hnRNPG from small nucleoplasmic granules to dense nuclear clusters (Figure S1C). Our chromatin preparation was treated with micrococcal nuclease, which degrades both DNA and RNA. Additional RNase treatment significantly increased the amount of co-precipitated RNAPII (Figure 1D), suggesting that the hnRNPG-RNAPII interaction is mediated through protein-protein interactions rather than through RNA. Our result is supported by an RNAPII CTD interactome study, which showed enrichment of hnRNPG with immunoprecipitation of S2P or S5P RNAPII (Nojima et al., 2018).

We examined the roles of different regions of hnRNPG in interacting with RNAPII. The hnRNPG protein consists of an RNA recognition motif (RRM) and a ~300-amino acid low-complexity sequence including two regions containing RGG motifs (RGG1 and RGG2). We introduced point mutations in the RRM, RGG1, or RGG2 region of hnRNPG to generate the mutants RRMmut, RGG1mut, and RGG2mut (Figure S1D), knocked down endogenous hnRNPG, and overexpressed FLAG-tagged forms of either wild type or mutant hnRNPG. Mutations in the RRM, RGG1, or RGG2 region decreased co-immunoprecipitation of RNAPII with hnRNPG (Figure 1E). The mutations in RRMmut are known to abolish RNA binding by the RRM region of hnRNPG (Moursy et al., 2014), therefore the RRM region might promote the interaction with RNAPII by enhancing binding of hnRNPG to nascent RNA at transcription sites. To test the possibility that m⁶A modifications in nascent RNA could influence the interaction of hnRNPG with transcribing RNAPII, we measured the interaction between hnRNPG and RNAPII upon knockdown of the m⁶A writer METTL3, which associates with RNAPII (Figure S1E) (Slobodin et al., 2017). METTL3 knockdown decreased the association of hnRNPG with RNAPII, but this effect was not significant (Figure 1F).

hnRNPG Binds Phosphorylated CTD of RNAPII and RNA Simultaneously

The RNAPII CTD is a docking site for many RNA processing factors and interacts with the low-complexity regions of multiple RNA-binding proteins (Burke et al., 2015; Harlen and Churchman, 2017; Kwon et al., 2013; Schwartz et al., 2013). We examined whether hnRNPG, which has an extensive low-complexity region (Figures S1D and S2A) (Dosztányi et al., 2005), can interact directly with the CTD. We used the GST-tagged CTD of RNAPII (GST-CTD) to pull down recombinant full-length hnRNPG protein (Figure S2B). Phosphorylated GST-CTD (GST-pCTD) was prepared by treatment with CDK7 or

CDK9 prior to the pull-down. Only the phosphorylated forms of GST-CTD were able to pull down hnRNPG (Figure 2A). Furthermore, pre-binding hnRNPG to a known RNA ligand, an A- or m⁶A-containing 34-nt hairpin from the long noncoding RNA metastasis-associated lung adenocarcinoma transcript 1 (MALAT1) (Liu et al., 2017), did not influence the pull-down of hnRNPG by GST-pCTD (Figure 2B). This result suggests that hnRNPG can simultaneously interact with RNA and pCTD. hnRNPG self-assembled *in vitro*, forming large assemblies with a radius of ~27 nm, with a slight increase in size in the presence of the MALAT1 RNA hairpin (Figure S2C). hnRNPG also formed large complexes on surface plasmon resonance (SPR) chips with immobilized MALAT1 RNA hairpin, evidenced by a continued slow increase after an initial rapid increase in SPR signal from RNA binding (Figure S2D). Addition of a synthetic peptide consisting of four heptapeptide repeats with phosphorylated Ser5 from the RNAPII CTD (S5P-peptide) reduced this larger complex formation without affecting initial RNA binding (Figure S2D). In solution, addition of S5P- or S2P-peptide had minimal effect on the size of hnRNPG assemblies measured by dynamic light scattering (Figure S2E).

The tendency of hnRNPG to form large assemblies complicated the quantitative measurement of RNA binding using native gel shift or nitrocellulose filters (not shown). To directly demonstrate RNA binding in the presence of GST-pCTD, we took advantage of the observation that hnRNPG remained soluble at 0.5 M NaCl but formed larger complexes upon transfer to 0.15–0.25 M NaCl that could be pelleted by centrifugation (spin-down assay). We found that adding S5P-peptide had minimal effect on binding of A- or m⁶A-containing MALAT1 RNA hairpin (Figure 2C), showing that hnRNPG can bind simultaneously to RNA and pCTD. We also used the spin-down assay to measure hnRNPG binding to pCTD. Both GST-CTD and GST-pCTD were soluble on their own (Figure S2F), but in the presence of hnRNPG, GST-pCTD co-precipitated with hnRNPG while unphosphorylated GST-CTD remained soluble (Figure 2D). The addition of MALAT1 RNA hairpin increased the fraction of hnRNPG found in the pellet but did not affect the co-precipitation of GST-pCTD (Figure 2D). These results validate that hnRNPG has much higher affinity for phosphorylated than for unphosphorylated CTD, and pCTD binding is unaffected by RNA binding.

The RRM, RGG1, and RGG2 Regions Mediate hnRNPG Binding to RNA and pCTD

To examine the effect of three functional domains in hnRNPG on RNA and CTD binding, we purified hnRNPG with mutations in the RRM, RGG1, or RGG2 region (Figures S1D and S2B). hnRNPG assembly size was not affected by mutations in RRM or RGG1 but slightly increased with mutations in the RGG2 region (Figure S2G). We used limited proteolysis to study hnRNPG binding to RNAs with distinct structures and lengths (Figure S2H). hnRNPG with or without equimolar RNA was treated with proteinase K, and its degradation was tracked over time (Figure 2E). Consistent with the preferential binding of hnRNPG to m⁶A-containing MALAT1 RNA hairpin (Liu et al., 2017), both MALAT1 RNAs protected hnRNPG from proteolysis, but the m⁶A-containing RNA had a more protective effect. This protection

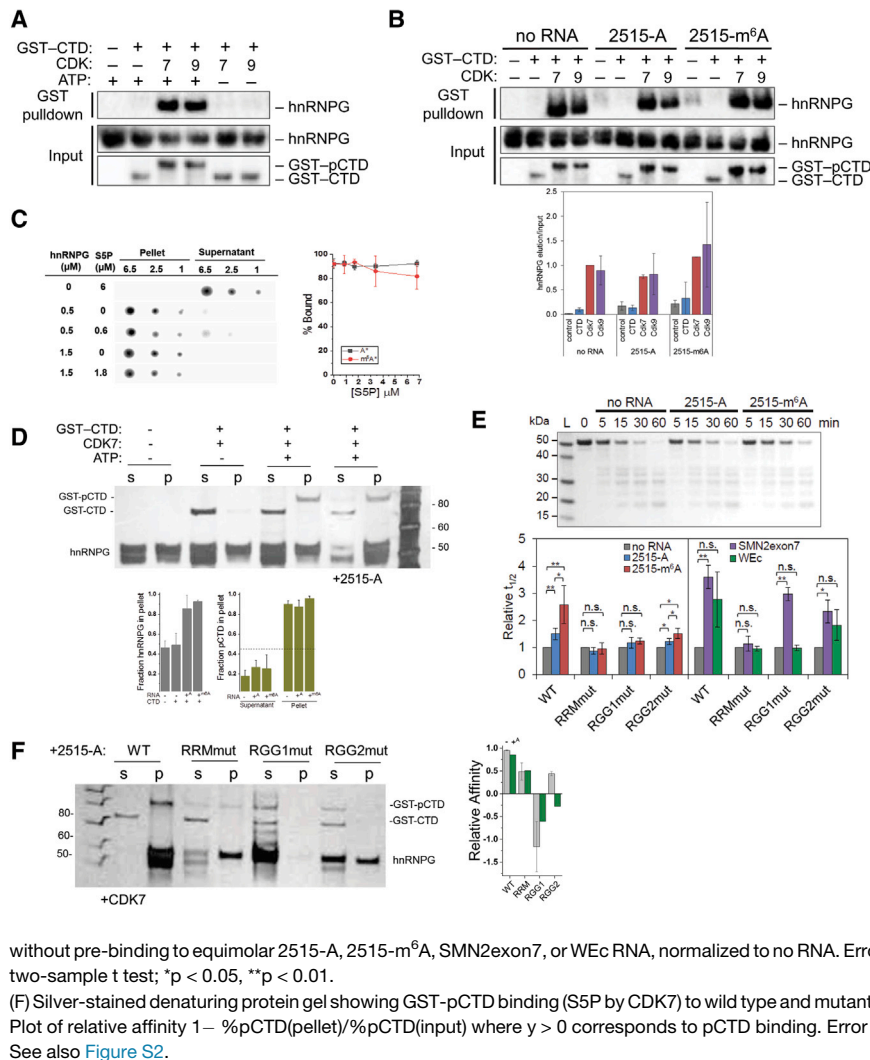


Figure 2. hnRNPG Binds Directly to Phosphorylated RNAPII CTD and Can Simultaneously Interact with Both RNA and CTD

(A) Western blot showing pull-down of recombinant hnRNPG with GST-tagged RNAPII CTD (GST-CTD) ± pre-treatment with CDK7, CDK9, and ATP. GST-pCTD, phosphorylated GST-CTD. Quantitation in (B).

(B) Western blot showing pull-down of recombinant hnRNPG ± MALAT1 RNA hairpin containing A (2515-A) or m⁶A (2515-m⁶A) with the GST-CTD ± phosphorylation by CDK7 or CDK9. Quantitation normalized to hnRNPG IP and to input: (RNAPII_{IP}/hnRNPG_{IP})/(RNAPII_{input}/hnRNPG_{input}). Error bars: ±1 SD; n = 3 replicates.

(C) Filter spotting showing RNA binding to recombinant hnRNPG by spin-down assay. Plot shows percent of bound MALAT1 RNA hairpin at varying concentrations of S5P-peptide. Error bars: ±1 SD; n = 3 independent experiments.

(D) Silver stained denaturing protein gel showing GST-pCTD binding (S5P by CDK7) to recombinant hnRNPG by spin-down assay. s, supernatant; p, pellet; +2515-A, with equimolar MALAT1 RNA hairpin. A and m⁶A superscripts correspond to addition of MALAT1 RNA with A or m⁶A. hnRNPG runs as two bands due to variable N-glycosylation. Fraction pCTD in the pellet refers to [GST-pCTD]/([GST-pCTD] + [GST-CTD]). Dashed line shows the fraction of hnRNPG in the pellet in the absence of GST-pCTD or RNA. Error bars: ±1 SD; n = 3 replicates.

(E) Denaturing protein gel showing time points from limited proteolysis of wild type hnRNPG ± MALAT1 RNA hairpin (2515-A/m⁶A) by proteinase K. Plot showing half-life (t_{1/2}) of wild type and mutant hnRNPG proteolysis by proteinase K, with or

without pre-binding to equimolar 2515-A, 2515-m⁶A, SMN2exon7, or Wec RNA, normalized to no RNA. Error bars, ±1 SD; n = 3–5 replicates; n.s., not significant by two-sample t test; *p < 0.05, **p < 0.01.

(F) Silver-stained denaturing protein gel showing GST-pCTD binding (S5P by CDK7) to wild type and mutant hnRNPG by spin-down assay. s, supernatant; p, pellet. Plot of relative affinity 1 – %pCTD(pellet)/%pCTD(input) where y > 0 corresponds to pCTD binding. Error bars: data variation; n = 2 independent experiments. See also Figure S2.

was reduced for RGG1mut and RGG2mut, and abolished for RRMmut. Mutations in the RRM also abolished the protective effect of two other RNAs bound by hnRNPG: unstructured RNA SMN2exon7 and structured RNA Wec (Kanhoush et al., 2010; Moursy et al., 2014). Mutations in RGG1 and RGG2 only slightly decreased the protective effect of SMN2exon7, which is bound by the RRM of hnRNPG (Moursy et al., 2014), but markedly blunted the protective effect of Wec, which is thought to bind the RGG2 region (Kanhoush et al., 2010). These results suggest that hnRNPG can bind both unstructured and structured RNA, with different roles for the RRM, RGG1 and RGG2 regions.

To examine the effect of domain mutations on RNAPII CTD binding, we first applied the GST-CTD pull-down assay at 0.5 M NaCl where the hnRNPG protein remained soluble (Figure S2I). Mutations in RRM reduced but did not abolish the interaction of hnRNPG with GST-CTD, whereas mutations in RGG1 strongly reduced and in RGG2 abolished this interaction. Next, we used our spin-down assay to measure pCTD binding to mutant hnRNPG proteins (Figures 2F, S2J, and S2K). Mutations in RGG1 or RGG2, but not in RRM, reduced pelleting of hnRNPG

compared to wild type. The RRM and RGG2 mutants, but not RGG1 mutant, could bind GST-pCTD in the absence of RNA (Figures S2J and S2K). On addition of MALAT1 RNA hairpin, RRM mutant remained capable of pCTD binding, but RGG2 mutant binding to pCTD was severely diminished (Figure 2F). In summary, we have used several independent *in vitro* assays to show that hnRNPG can simultaneously bind RNA and RNAPII pCTD. The RRM and two RGG regions all function in RNA binding, while the RGG regions are required for the direct interaction of hnRNPG with pCTD.

The RRM, RGG1, and RGG2 Regions of hnRNPG Affect Transcript Abundance

To examine the effect of the RRM, RGG1, and RGG2 regions on the cellular function of hnRNPG, we knocked down endogenous hnRNPG and overexpressed either negative control vector coding only for a FLAG tag (NCV), wild type FLAG-hnRNPG (WT), or mutant FLAG-hnRNPG with mutations in the RRM, RGG1, or RGG2 region (RRMmut, RGG1mut, or RGG2mut) (Figures S1D and S3A). We conducted mRNA sequencing (mRNA-seq) to

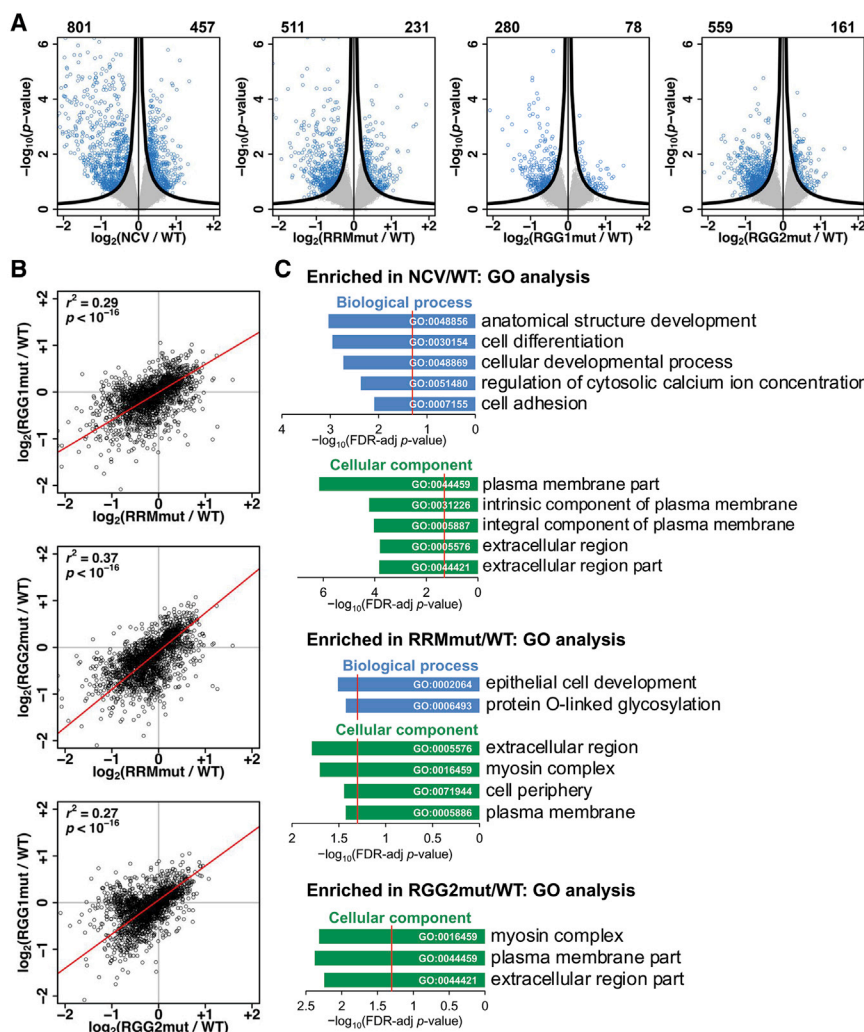


Figure 3. The RRM, RGG1, and RGG2 Regions Function in Regulation of Transcript Abundance by hnRNP G

(A) Volcano plots showing $\log_2(\text{fold change})$ and $-\log_{10}(p\text{-value})$ for changes in gene expression in NCV, RRMmut, RGG1mut, and RGG2mut relative to WT. Genes satisfying the π value threshold (see STAR Methods) were considered differentially expressed. Numbers of down- and upregulated genes are listed at the top left and right sides, respectively, of each plot. Black curves, π value threshold; blue points, differentially expressed exons; gray points, non-differentially expressed exons.

(B) Correlated changes in gene expression, quantified as $\log_2(\text{fold change relative to WT})$, in mRNA sequencing data for RRMmut, RGG1mut, and RGG2mut. Each point is a gene. r , Pearson correlation coefficient; p , p value using Fisher transformation; red line, model II major axis linear regression.

(C) Gene ontology (GO) analysis showing false discovery rate (FDR) adjusted p value ($-\log_{10}$) for biological processes (blue) and cellular components (green) enriched among genes differentially expressed relative to WT. Red line, FDR threshold ($p = 0.05$). Number (n) of genes in each category from top to bottom: NCV/WT biological process ($n = 295, 214, 215, 32, 68$), cellular component ($n = 186, 122, 116, 275, 236$); RRMmut/WT biological process ($n = 18, 13$), cellular component ($n = 156, 9, 176, 170$); RGG2mut/WT cellular component ($n = 10, 97, 130$).

See also Figure S3 and Table S1.

detect changes in transcript abundance relative to wild type (Table S1), and we validated these changes by real-time qPCR (Figures S3B and S3C). Relative to cells expressing wild type FLAG-hnRNP G, 1,258 genes were differentially expressed in NCV, and 300–800 genes were differentially expressed for each hnRNP G mutant (Figure 3A). Changes in transcript abundance were correlated among the three mutants and NCV (Figures 3B and S3D), indicating that the RRM, RGG1, and RGG2 regions all contribute to the regulation of transcript abundance, and mutations in these regions led to partial loss-of-function. Differentially expressed genes were enriched for functions in developmental processes (Figure 3C), consistent with known functions of hnRNP G in development (Dichmann et al., 2008; Tsend-Ayush et al., 2005).

The RRM, RGG1, and RGG2 Regions of hnRNP G Function in Alternative Splicing Regulation

Using published mRNA sequencing data (Liu et al., 2017), we identified exons that were differentially expressed upon hnRNP G, METTL3, or METTL14 knockdown (regulated exons), or showed correlated changes in expression upon hnRNP G

and either METTL3 or METTL14 knock-down (co-regulated exons) (Figures 4A and 4B). We found that 18,031 exons were either co-downregulated or co-upregulated by hnRNP G and METTL3/14 knockdown, indicating that hnRNP G and METTL3/14 in wild type cells promote their inclusion or exclusion, respectively (Figure 4B). Exons regulated by hnRNP G, or co-regulated by hnRNP G and METTL3/14, were enriched in genes that function in metabolic processes and RNA binding (Figures 4C and S4A). Using previously published photoactivatable ribonucleoside-enhanced crosslinking and immunoprecipitation followed by methylated RNA immunoprecipitation (PAR-CLIP-MeRIP) sequencing data (Liu et al., 2017), we examined the distribution of hnRNP G-bound m⁶A sites around hnRNP G- and METTL3/14-regulated exons (Figures 4D and S4B). We have previously shown that 81% of hnRNP G-bound m⁶A sites identified by hnRNP G PAR-CLIP-MeRIP were also m⁶A methyltransferase-dependent (Liu et al., 2017). We found that hnRNP G-bound m⁶A sites were enriched in exonic regions near splice sites (Figure 4D), similar to the known pattern of m⁶A sites in nascent mRNA (Louloupi et al., 2018). Co-downregulated exons were more likely to have hnRNP G-bound m⁶A sites near their splice sites than co-upregulated exons (Figures 4D and S4C), suggesting that m⁶A plays a bigger role in hnRNP G-dependent exon inclusion.

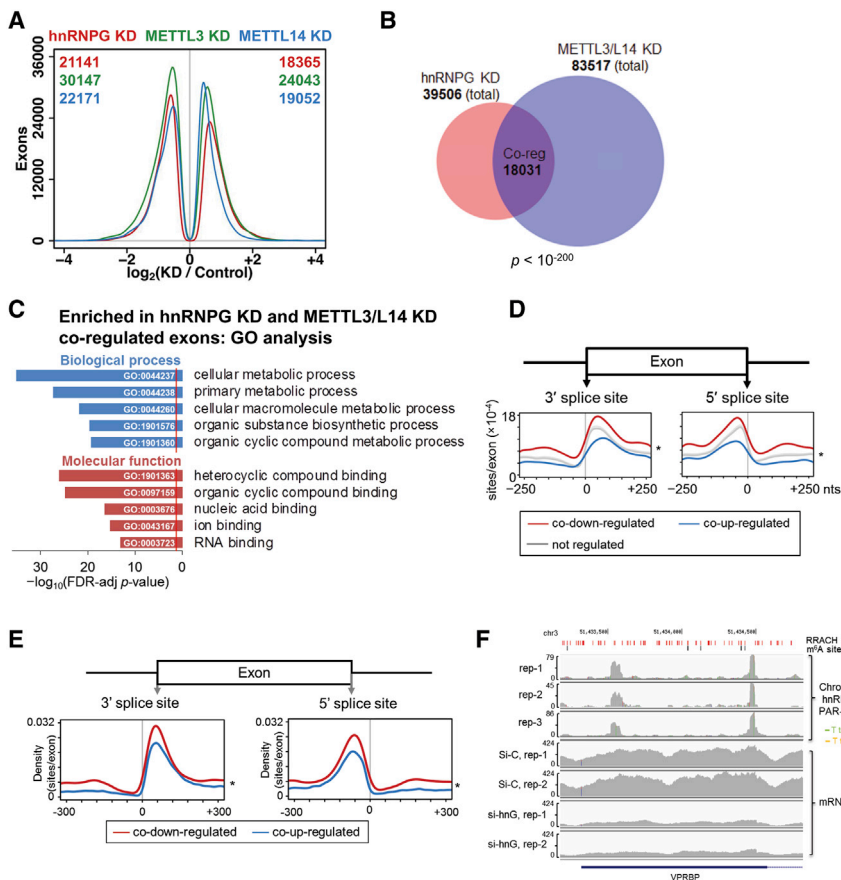


Figure 4. hnRNPG and m⁶A Methyltransferase Co-regulate Exon Splicing

(A) Distribution of differentially expressed exons showing number of exons at each log₂(fold change) for hnRNPG (red), METTL3 (green), and METTL14 (blue) knockdown relative to control knockdown (Liu et al., 2017). Numbers of down- and upregulated differentially expressed exons for each knockdown are listed at the top left and right corners of the plot in the color of the knockdown. (B) Venn diagram showing overlap between exons differentially expressed upon hnRNPG knockdown (39,506) and upon METTL3 or METTL14 knockdown (83,517). Overlapping exons are co-regulated exons (Co-reg, 18,031), with correlated changes in expression upon hnRNPG KD and either METTL3 or METTL14 KD.

(C) GO analysis showing false discovery rate (FDR) adjusted p value ($-\log_{10}$) for biological processes (blue) and molecular functions (red) enriched among genes containing differentially expressed exons that were co-regulated by hnRNPG KD and either METTL3 or METTL14 KD. Red line, FDR threshold ($p = 0.05$).

(D) hnRNPG-bound m⁶A sites per regulated exon at each site in the -250 to $+250$ nt region around the splice sites of exons co-downregulated (11,996), co-upregulated (9,818), or not regulated (140,932) upon hnRNPG KD and either METTL3 or METTL14 KD. $*p < 10^{-16}$ by paired t test between curves for down- versus upregulated exons.

(E) Same as (D) but using hnRNPG binding sites from chromatin associated PAR-CLIP data.

(F) Sequencing track of VPRBP gene showing read density by chromatin-associated hnRNPG PAR-CLIP. T-to-C and T-to-A mutations (marks of

cross-linking in PAR-CLIP) are shown as green and orange vertical lines. rep, biological replicate; si-C, control siRNA, si-hnG: hnRNPG siRNA. RRACH motifs, red; m⁶A sites, black.

See also Figure S4 and Table S2.

Even though hnRNPG is strictly localized to the nucleus (Figure S1C), the above analysis would not distinguish between chromatin-associated nascent RNAs and nucleoplasmic RNAs bound by hnRNPG. Therefore, we performed hnRNPG PAR-CLIP on the chromatin fraction to sequence only chromatin-bound mRNAs crosslinked to hnRNPG (Figures S4D and S4E; Table S2). Similar to hnRNPG-bound m⁶A sites in whole cell RNA (Figure 4D), hnRNPG-bound sites in chromatin-bound RNA (Figures 4E and 4F) were enriched in exonic regions near the splice sites of exons co-regulated by hnRNPG and METTL3/14, and this enrichment was stronger in co-downregulated than in co-upregulated exons. The contrast between densities on the exonic versus intronic side of splice sites was even more pronounced for hnRNPG-bound sites in chromatin-bound RNAs (Figure 4E) than in whole cell RNAs (Figure 4D). Thus, hnRNPG binds to chromatin-associated nascent RNAs in a specific pattern near the splice sites of co-regulated exons, and this pattern is more prominent in nascent RNA than in whole-cell RNA, consistent with our model that hnRNPG-mediated alternative splicing primarily occurs co-transcriptionally.

The m⁶A reader YTHDC1 has been implicated in m⁶A-dependent regulation of alternative splicing (Xiao et al., 2016). Using published sequencing data (Liu et al., 2017; Xiao et al., 2016),

we found that 6,669 exons were differentially expressed upon both YTHDC1 and hnRNPG knockdown (Figure S4F), with similar numbers of co-regulated and anti-co-regulated exons (Figure S4G). The m⁶A installation motif RRACH was enriched in exonic regions for both hnRNPG- and YTHDC1-regulated exons, but this enrichment was more pronounced in hnRNPG-regulated exons (Figure S4H).

We also analyzed our mRNA-seq data (Table S1) for changes in exon splicing in cells expressing NCV, RRMmut, RGG1mut, or RGG2mut relative to cells expressing wild type hnRNPG (WT). We found thousands of differentially expressed exons for each mutant and NCV (Figures 5A–5C), which were validated by RT-PCR (Figure S5A). Mutations in RGG1 affected fewer exons compared to mutations in RRM or RGG2, and mutations in RRM and RGG2 led to many more downregulated than upregulated exons, suggesting that these regions mainly promote exon inclusion in wild type cells (Figures 5A and 5B). Changes in exon expression were correlated among cells expressing NCV and the three hnRNPG mutants (Figures 5D and S5B). The correlation between RRMmut and RGG2mut was much stronger than the correlation of either mutant with RGG1mut, possibly due to overlapping functions of RRM and RGG2 in nascent RNA binding. Genes containing exons regulated by NCV, RRMmut, and

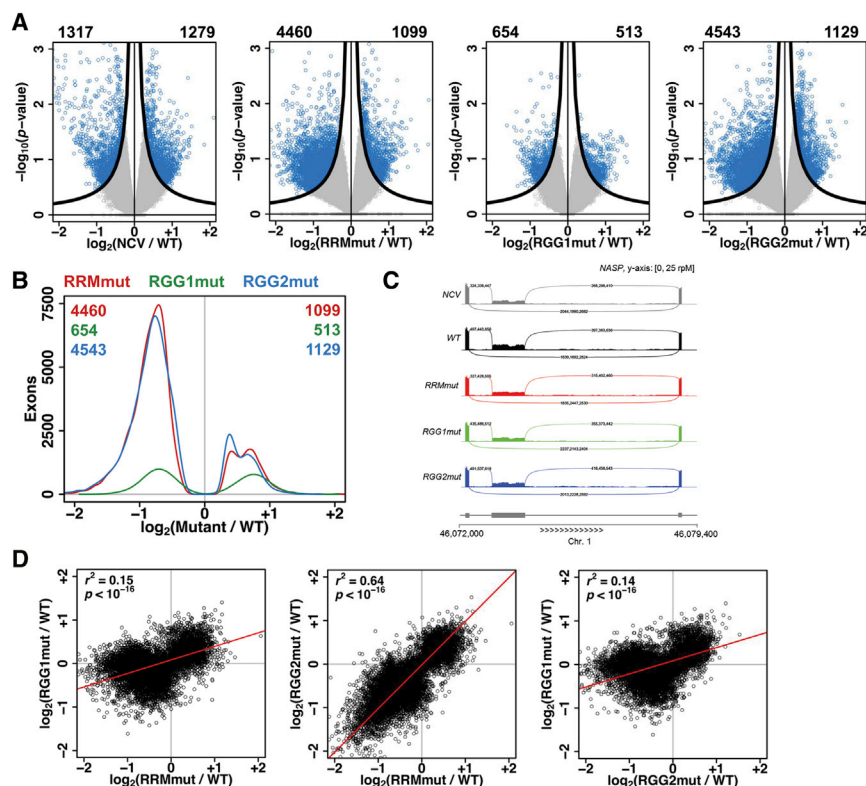


Figure 5. The RRM, RGG1, and RGG2 Regions Function in Regulation of Alternative Splicing by hnRNP G

(A) Volcano plots showing $\log_2(\text{fold change})$ and $-\log_{10}(p\text{-value})$ for changes in exon expression in NCV, RRMmut, RGG1mut, and RGG2mut relative to WT. Exons satisfying the π value threshold (see STAR Methods) were considered differentially expressed. Numbers of down- and upregulated differentially expressed exons are listed at the top left and right corners of each plot. Black curves, π value threshold; blue points, differentially expressed exons; gray points, non-differentially expressed exons.

(B) Distribution of differentially expressed exons showing number of exons at each $\log_2(\text{fold change})$ for RRMmut (red), RGG1mut (green), and RGG2mut (blue) relative to WT. Numbers of down- and upregulated differentially expressed exons for each mutant are listed at the top left and top right corners of the plot, in the color of the mutant.

(C) mRNA-seq reads for NASP transcripts in NCV (gray), WT (black), RRMmut (red), RGG1mut (green), and RGG2mut (blue) expressing cells. Sashimi plot shows downregulation of the middle exon in NCV, RRMmut, RGG1mut, and RGG2mut compared to WT cells.

(D) Correlated changes in exon expression, quantified as $\log_2(\text{fold change relative to WT})$, in mRNA sequencing data for RRMmut, RGG1mut, and RGG2mut. Each point is an exon. r , Pearson correlation coefficient; p , p value using Fisher transformation; red line, model II major axis linear regression.

See also Figures S5 and S6 and Table S1.

RRG1mut were enriched for biological processes in metabolism and development (Figure S5C). Unexpectedly, more exons were downregulated in cells expressing the RRM and RRG2 mutants ($\sim 4,500$) than in cells expressing NCV ($\sim 1,300$) (Figure 5A). Because RRM and RGG2 mutations decrease RNA binding (Figure 2E), this result is unlikely from off-target binding and more likely from a dominant negative effect in which RRM and RGG2 mutants lose RNA binding but still associate with pCTD to block its interactions with other proteins. In contrast, in cells expressing NCV, the simultaneous loss of interactions of hnRNP G with both nascent RNA and RNAPII may have a smaller effect on splicing because the CTD retains its interactions with other proteins.

m⁶A Site Position in the Regulation of Alternative Splicing by hnRNP G

To further study the roles of RRM, RGG1, and RGG2 in the m⁶A-dependent regulation of alternative splicing by hnRNP G, we examined the pattern of hnRNP G-bound m⁶A sites around the splice sites of hnRNP G mutant-regulated exons. hnRNP G-bound m⁶A sites were more enriched in exonic regions near splice sites of exons downregulated upon hnRNP G knockdown (NCV) than in upregulated exons (Figure 6A). Similar enrichment was observed at exons regulated by hnRNP G mutants, however, this enrichment was less prominent at RGG1mut-regulated exons than at RRMmut- or RGG2mut-regulated exons (Figures

6A and 6B). hnRNP G-bound sites in nascent mRNA were also enriched in exonic regions near splice sites, and the contrast between site densities in exonic and intronic regions was even more pronounced than in whole-cell mRNA (Figure 6C). NCV- and RGG1mut-regulated exons in nascent RNA had higher densities of hnRNP G-bound sites for down- versus upregulated exons at both splice sites, while RRMmut- and RGG2mut-regulated exons had higher or equal densities of hnRNP G-bound sites for up- versus downregulated exons at the 3' or 5' splice site, respectively. A particularly pronounced effect of RRM and RGG2 mutation was the reduction of the density of the hnRNP G-bound m⁶A sites at downregulated exons, consistent with the observed reduction in exon inclusion for these two mutants (Figure 5B). These results suggest that RRM and RGG2 regions of hnRNP G have overlapping functions in nascent RNA binding and regulation of exon inclusion.

A role for RGG2 in m⁶A-dependent splicing regulation would be consistent with the function of this region in the preferential binding of m⁶A-modified RNAs by hnRNP G (Liu et al., 2017). Genes containing RGG2mut-regulated exons were enriched in metabolic functions (Figure S6A), similar to exons co-regulated by hnRNP G and m⁶A methyltransferase (Figure 4C), and in functions in nervous system development, consistent with the known role of hnRNP G in neural development (Dichmann et al., 2008; Shashi et al., 2015; Tsend-Ayush et al., 2005). Further supporting the importance of RGG2 in splicing

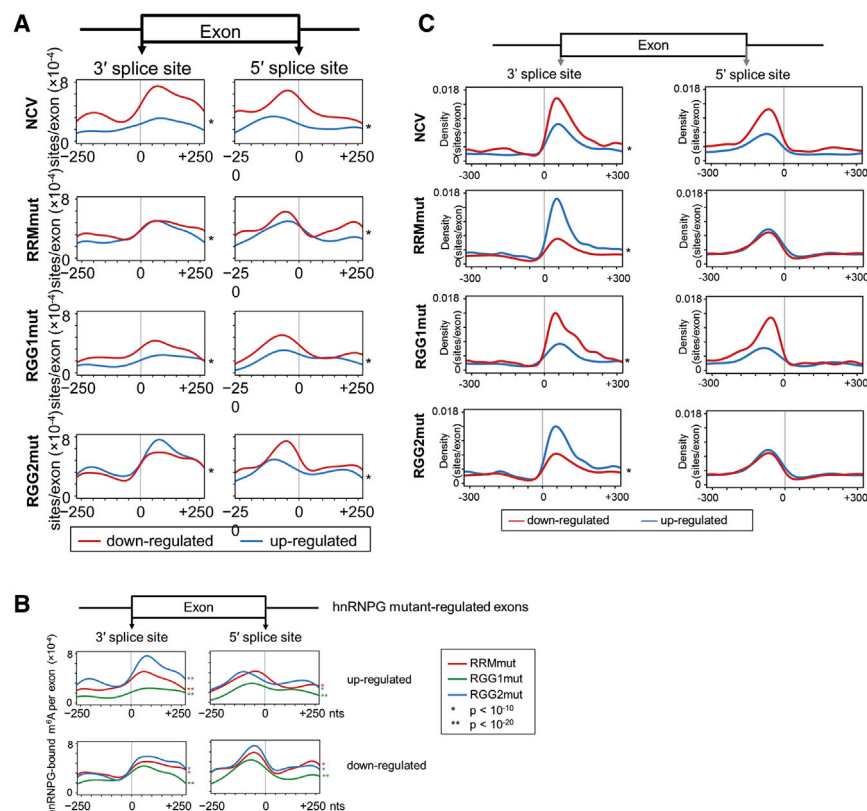


Figure 6. Role of m⁶A Site Position in Regulation of Alternative Splicing by hnRNPG

(A) Distribution of hnRNPG-bound m⁶A sites per regulated exon at each site in the -250 to +250 nt region around the splice sites of exons down-regulated (red) or up-regulated (blue) in NCV (1,317 down, 1,279 up), RRMmut (4,460 down, 1,099 up), RGG1mut (654 down, 513 up), and RGG2mut (4,543 down, 1,129 up) relative to WT. *p < 10⁻¹⁶ by paired t test between curves for down- versus up-regulated exons.

(B) Overlay curves from (A) for RRMmut, RGG1mut, and RGG2mut.

(C) Same as (A) but using hnRNPG binding sites from chromatin-associated PAR-CLIP data.

See also Figure S6.

knockdown significantly reduced RNAPII density near the splice sites of co-down-regulated exons (Figures 7A–7C). These results support an association between RNAPII occupancy and hnRNPG-mediated regulation of exon inclusion. The RNAPII density for co-upregulated exons was similar to that for non-regulated exons in magnitude and shape, suggesting that hnRNPG dependent regulation of exon exclusion is at best only weakly associated with RNAPII occupancy.

We propose a model in which hnRNPG assemblies interact co-transcriptionally with both the m⁶A sites in nascent mRNA and the CTD of transcribing RNAPII to regulate exon inclusion (Figure 7E). Our model assumes that m⁶A modifications are installed co-transcriptionally shortly after nascent RNA emerges from the exit channel of RNAPII, consistent with literature reports (Barbieri et al., 2017; Huang et al., 2019; Ke et al., 2017; Knuckles et al., 2017; Louloui et al., 2018; Xiang et al., 2017). A crucial feature is the ability of hnRNPG to assemble into a large complex, which allows simultaneous interactions with nascent mRNA, RNAPII CTD, and m⁶A. We propose the following order of events for hnRNPG- and m⁶A-dependent exon inclusion: RNAPII transcribes through the exon-intron junction; m⁶A is installed; hnRNPG binds to the m⁶A site, which is kinetically favored through its association with RNAPII CTD; RNAPII then dwells longer near the exon-intron junction; and increased RNAPII dwell time increases splice site utilization, perhaps through recruitment of spliceosomal components or splicing factors.

DISCUSSION

In this study, we showed that the RNA-binding protein hnRNPG uses RGG regions to directly interact with the phosphorylated CTD of RNAPII. hnRNPG binding to RNA and the RNAPII CTD can occur simultaneously, likely by assembly of hnRNPG into large complexes. In cells, the RRM, RGG1, and RGG2 regions of hnRNPG functioned in the regulation of alternative splicing. We further found that hnRNPG-bound m⁶A sites near the splice

regulation, cells expressing RGG2mut showed significantly decreased proliferation compared to cells expressing wild type hnRNPG (Figure S6B).

m⁶A-Mediated Regulation of RNAPII Occupancy and Alternative Splicing by hnRNPG

RNAPII dwell time in transcription elongation can influence exon usage by enhancing or blocking recognition of a splice site by splicing factors (Kornblihtt et al., 2013). To study the association between m⁶A modification, hnRNPG binding, and RNAPII occupancy at relevant exons, we knocked down hnRNPG (Figure S7A) (Liu et al., 2017) and performed quantitative chromatin-immunoprecipitation followed by sequencing (ChIP-seq) (Figures 7A–7D and S7B; Table S3) using antibodies for total, S2P, and S5P RNAPII. Each sample was spiked with a known amount of *Drosophila* chromatin for cross-sample comparison of RNAPII densities (Egan et al., 2016). After normalization to the spiked-in *Drosophila* chromatin, the overall RNAPII ChIP-seq density across all gene regions decreased upon hnRNPG knockdown (Figure S7C). To identify the effects of hnRNPG on splicing, we focused our next analysis on regions of hnRNPG-dependent alternative splicing regulation. The total, S2P, and S5P RNAPII densities were analyzed separately for exons that are co-downregulated, co-upregulated, and non-regulated upon hnRNPG or METTL3/14 knockdown. In the control samples, the densities of total, S2P, and S5P RNAPII were much higher for co-downregulated than for co-upregulated or non-regulated exons, with a peak in RNAPII density ~80 nt downstream of the 3' splice site (Figures 7A–7C). Moreover, hnRNPG

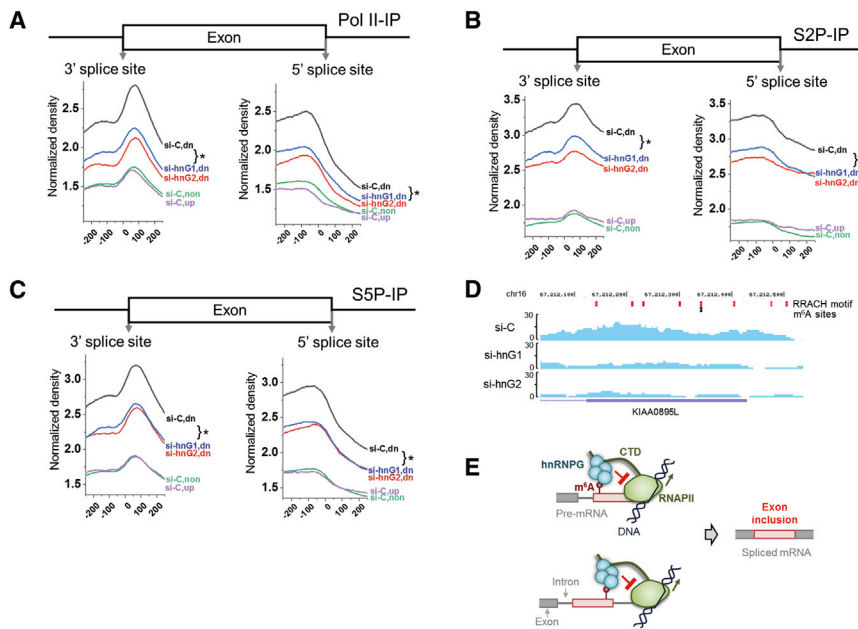


Figure 7. m⁶A-Dependent Regulation of Alternative Splicing by hnRNPG and RNAPII Occupancy

(A) RNAPII ChIP-seq upon hnRNPG knockdown, using total RNAPII antibody. si-C, control knockdown; hnG1 and hnG2, hnRNPG knockdown with two siRNAs. Co-regulated exons downregulated (dn, 15,318), upregulated (up, 11,255), and non-regulated (non, 35,377) upon hnRNPG or METTL3/14 knockdown. * $p < 10^{-4}$.

(B) Same as (A) using S2P-CTD antibody.

(C) Same as (A) using S5P-CTD antibody.

(D) ChIP-seq tracks from KIAA0895L gene using RNAPII-S2P antibody in cells treated with control or hnRNPG siRNA. si-C, control siRNA; si-hnG1 and si-hnG2, hnRNPG knockdown with two siRNAs. RRACH motifs, red; m⁶A sites, black.

(E) Model for m⁶A-dependent regulation of exon inclusion by hnRNPG. Red circle, m⁶A site; red box, alternative exon; gray, constitutive exon; black lines, DNA; green, RNAPII with CTD as extended line; cyan, hnRNPG complex. RNAPII transcribes through the splice site, m⁶A is installed, and hnRNPG interacts with m⁶A, causing RNAPII to increase dwell time downstream of the m⁶A site, which results in hnRNPG- and m⁶A-dependent exon inclusion.

See also Figure S7 and Table S3.

sites of regulated exons were associated with increased RNAPII occupancy and exon inclusion. Our results support a model in which hnRNPG assemblies interact co-transcriptionally with nascent RNA and the RNAPII CTD, while m⁶A sites in nascent RNA promote hnRNPG binding to modulate RNAPII occupancy and alternative splicing.

One possible mechanism for hnRNPG- and m⁶A-dependent alternative splicing regulation is through RNAPII pausing to provide an appropriate time window and spatial resolution to recruit splicing factors. RNAPII pausing during transcription elongation can be modulated by RNA sequence and structure, chromatin state, and RNA-protein interactions (Mayer et al., 2017). Although RNAPII density by ChIP-seq could reflect RNAPII pausing, ChIP-seq has limited resolution (100–200 nt) and cannot distinguish between RNAPII pausing and other etiologies of RNAPII occupancy. A future direction could utilize the native elongating transcript sequencing (NET-seq) method to more accurately measure RNAPII pausing (Mayer et al., 2015; Nojima et al., 2015) and m⁶A-dependent splicing regulation. These published NET-seq results have shown that RNAPII density is higher in exons than in introns and is also high around splice sites. NET-seq upon the depletion of hnRNPG, METTL3, and/or METTL14 could provide the single-base resolution needed to firmly establish the potential role of RNAPII pausing in m⁶A-dependent exon inclusion.

The RNAPII CTD coordinates co-transcriptional RNA processing with transcription by recruiting constitutive RNA processing factors through direct interactions. However, the role of the CTD in recruiting alternative splicing factors, particularly through direct interactions, is less clear (Bentley, 2014). Our study demonstrates that an alternative splicing factor, hnRNPG, directly interacts with the RNAPII CTD. The selective interaction

of hnRNPG with phosphorylated CTD and the effect of transcription inhibitors on the hnRNPG-RNAPII interaction support our conclusion that hnRNPG interacts with RNAPII co-transcriptionally. We also found that the direct interaction of hnRNPG with phosphorylated CTD depended on two RGG regions in its low-complexity region. To our knowledge, RGG regions have not previously been shown to directly bind the CTD. Because RGG motifs are the second most common RNA binding motif among mammalian mRNA-binding proteins (Gerstberger et al., 2014), our findings suggest that many other RNA-binding proteins might regulate pre-mRNA processing by interacting directly with the RNAPII CTD through their RGG regions. Moreover, RGG motifs are commonly found in low-complexity regions, so a direct interaction between RGG motifs and the CTD could also have implications for low-complexity region assembly. The CTD is known to interact with low-complexity regions of several RNA-binding proteins, and these interactions have proposed functions in transcriptional regulation (Harlen and Churchman, 2017; Hnisz et al., 2017). hnRNPG forms a distinct interaction with the RNAPII CTD for co-transcriptional splicing regulation.

The abundant m⁶A modification in mRNA has been implicated in alternative splicing regulation. Most m⁶A modifications are deposited co-transcriptionally and enriched in exonic regions near splice sites (Barbieri et al., 2017; Ke et al., 2017; Knuckles et al., 2017; Louloui et al., 2018), but it is unknown how m⁶A near splice sites can modulate co-transcriptional alternative splicing. Because alternative splicing regulation by RNA-binding proteins depends on their binding site positions (Fu and Ares, 2014), m⁶A site positions, which determine m⁶A reader protein binding sites, likely also influence alternative splicing regulation. We found that hnRNPG-regulated exons were associated with

specific patterns of hnRNPG-bound m⁶A sites and RNAPII occupancy, supporting a model in which hnRNPG binds m⁶A sites near target exons and modulates RNAPII occupancy to promote exon inclusion. The regulation of alternative splicing involves a complex “splicing code” of *cis*- and *trans*-acting factors (Fu and Ares, 2014). We have shown that a *cis*-acting RNA modification, m⁶A, modulates the regulation of alternative splicing by hnRNPG by recruiting hnRNPG to specific sites, while a *trans*-acting protein, hnRNPG, interacts with the transcription machinery and co-transcriptionally modulates RNAPII occupancy to regulate alternative splicing. The m⁶A-dependent regulation of alternative splicing by hnRNPG demonstrates how the RNAPII CTD, RGG regions, and m⁶A act together to modulate the co-transcriptional regulation of alternative splicing by a low-complexity m⁶A reader protein.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.molcel.2019.07.005>.

ACKNOWLEDGMENTS

We thank all members of the Pan lab and Dr. Jingyi Fei for helpful discussion. We thank Dr. Qing Dai for chemical synthesis of RNA oligonucleotides, and

Drs. Gabriel Salzman and Demet Araç-Özkan for guidance with protein purification. We thank the University of Chicago BioPhysics Core Facility, Integrated Light Microscopy Core Facility, and Genomics Facility for assistance with biophysics, microscopy, and sequencing experiments. This work was supported by the NIH (R01GM113194 and RM1HG008935 to T.P. and F30GM120917 to K.I.Z.), the National Institutes of Health Medical Scientist Training Program (T32GM007281), and the University of Chicago Biological Sciences Division and Frank Family Endowment (K.I.Z.).

AUTHOR CONTRIBUTIONS

Conceptualization, K.I.Z. and T.P.; Methodology, K.I.Z., H.S., and T.P.; Software, M.P. and R.L.; Formal Analysis, K.I.Z., R.L., and M.P.; Investigation, K.I.Z., H.S., A.C.W., Ž.M., and J.N.P.; Writing – Original Draft, K.I.Z.; Writing – Review & Editing, K.I.Z., H.S., C.H., M.P., and T.P.; Supervision, C.H. and T.P.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: July 24, 2018

Revised: May 20, 2019

Accepted: July 2, 2019

Published: August 21, 2019

REFERENCES

- Alarcón, C.R., Goodarzi, H., Lee, H., Liu, X., Tavazoie, S., and Tavazoie, S.F. (2015). HNRNPA2B1 Is a Mediator of m(6)A-Dependent Nuclear RNA Processing Events. *Cell* 162, 1299–1308.
- Araç, D., Boucard, A.A., Bolliger, M.F., Nguyen, J., Soltis, S.M., Südhof, T.C., and Brunker, A.T. (2012). A novel evolutionarily conserved domain of cell-adhesion GPCRs mediates autoprolysis. *EMBO J.* 31, 1364–1378.
- Ashburner, M., Ball, C.A., Blake, J.A., Botstein, D., Butler, H., Cherry, J.M., Davis, A.P., Dolinski, K., Dwight, S.S., Eppig, J.T., et al.; The Gene Ontology Consortium (2000). Gene ontology: tool for the unification of biology. *Nat. Genet.* 25, 25–29.
- Barbieri, I., Tzelepis, K., Pandolfini, L., Shi, J., Millán-Zambrano, G., Robson, S.C., Aspris, D., Migliori, V., Bannister, A.J., Han, N., et al. (2017). Promoter-bound METTL3 maintains myeloid leukaemia by m⁶A-dependent translation control. *Nature* 552, 126–131.
- Bartosovic, M., Molares, H.C., Gregorova, P., Hrossova, D., Kudla, G., and Vanacova, S. (2017). N6-methyladenosine demethylase FTO targets pre-mRNAs and regulates alternative splicing and 3'-end processing. *Nucleic Acids Res.* 45, 11356–11370.
- Bentley, D.L. (2014). Coupling mRNA processing with transcription in time and space. *Nat. Rev. Genet.* 15, 163–175.
- Borsos, B.N., Huliák, I., Majoros, H., Ujfaludi, Z., Gyenis, Á., Pukler, P., Boros, I.M., and Pankotai, T. (2017). Human p53 interacts with the elongating RNAPII complex and is required for the release of actinomycin D induced transcription blockage. *Sci. Rep.* 7, 40960.
- Brugiolo, M., Botti, V., Liu, N., Müller-McNicol, M., and Neugebauer, K.M. (2017). Fractionation iCLIP detects persistent SR protein binding to conserved, retained introns in chromatin, nucleoplasm and cytoplasm. *Nucleic Acids Res.* 45, 10452–10465.
- Burke, K.A., Janke, A.M., Rhine, C.L., and Fawzi, N.L. (2015). Residue-by-Residue View of In Vitro FUS Granules that Bind the C-Terminal Domain of RNA Polymerase II. *Mol. Cell* 60, 231–241.
- Corcoran, D.L., Georgiev, S., Mukherjee, N., Gottwein, E., Skalsky, R.L., Keene, J.D., and Ohler, U. (2011). PARalyzer: definition of RNA binding sites from PAR-CLIP short-read sequence data. *Genome Biol.* 12, R79.
- Dai, Q., Fong, R., Saikia, M., Stephenson, D., Yu, Y.T., Pan, T., and Piccirilli, J.A. (2007). Identification of recognition residues for ligation-based detection and quantitation of pseudouridine and N6-methyladenosine. *Nucleic Acids Res.* 35, 6322–6329.

- Dichmann, D.S., Fletcher, R.B., and Harland, R.M. (2008). Expression cloning in *Xenopus* identifies RNA-binding proteins as regulators of embryogenesis and *Rbm* as necessary for neural and muscle development. *Dev. Dyn.* 237, 1755–1766.
- Dobin, A., Davis, C.A., Schlesinger, F., Drenkow, J., Zaleski, C., Jha, S., Batut, P., Chaisson, M., and Gingeras, T.R. (2013). STAR: ultrafast universal RNA-seq aligner. *Bioinformatics* 29, 15–21.
- Dominissini, D., Moshitch-Moshkovitz, S., Schwartz, S., Salmon-Divon, M., Ungar, L., Osenberg, S., Cesarkas, K., Jacob-Hirsch, J., Amariglio, N., Kupiec, M., et al. (2012). Topology of the human and mouse m6A RNA methylomes revealed by m6A-seq. *Nature* 485, 201–206.
- Dosztányi, Z., Csizmek, V., Tompa, P., and Simon, I. (2005). IUPred: web server for the prediction of intrinsically unstructured regions of proteins based on estimated energy content. *Bioinformatics* 21, 3433–3434.
- Egan, B., Yuan, C.-C., Craske, M.L., Labhart, P., Guler, G.D., Arnott, D., Maile, T.M., Busby, J., Henry, C., Kelly, T.K., et al. (2016). An Alternative Approach to ChIP-Seq Normalization Enables Detection of Genome-Wide Changes in Histone H3 Lysine 27 Trimethylation upon EZH2 Inhibition. *PLoS ONE* 11, e0166438.
- Ferrer, C.M., Alders, M., Postma, A.V., Park, S., Klein, M.A., Cetinbas, M., Pajkrt, E., Glas, A., van Koningsbruggen, S., Christoffels, V.M., et al. (2018). An inactivating mutation in the histone deacetylase SIRT6 causes human perinatal lethality. *Genes Dev.* 32, 373–388.
- Fu, X.-D., and Ares, M., Jr. (2014). Context-dependent control of alternative splicing by RNA-binding proteins. *Nat. Rev. Genet.* 15, 689–701.
- Gerstberger, S., Hafner, M., and Tuschl, T. (2014). A census of human RNA-binding proteins. *Nat. Rev. Genet.* 15, 829–845.
- Groisman, R., Polanowska, J., Kuraoka, I., Sawada, J., Saijo, M., Drapkin, R., Kisselev, A.F., Tanaka, K., and Nakatani, Y. (2003). The ubiquitin ligase activity in the DDB2 and CSA complexes is differentially regulated by the COP9 signalosome in response to DNA damage. *Cell* 113, 357–367.
- Harlen, K.M., and Churchman, L.S. (2017). The code and beyond: transcription regulation by the RNA polymerase II carboxy-terminal domain. *Nat. Rev. Mol. Cell Biol.* 18, 263–273.
- Hnisz, D., Shrinivas, K., Young, R.A., Chakraborty, A.K., and Sharp, P.A. (2017). A Phase Separation Model for Transcriptional Control. *Cell* 169, 13–23.
- Hofmann, Y., and Wirth, B. (2002). hnRNP-G promotes exon 7 inclusion of survival motor neuron (SMN) via direct interaction with Htra2- β 1. *Hum. Mol. Genet.* 11, 2037–2049.
- Hsin, J.-P., and Manley, J.L. (2012). The RNA polymerase II CTD coordinates transcription and RNA processing. *Genes Dev.* 26, 2119–2137.
- Huang, H., Weng, H., Zhou, K., Wu, T., Zhao, B.S., Sun, M., Chen, Z., Deng, X., Xiao, G., Auer, F., et al. (2019). Histone H3 trimethylation at lysine 36 guides m⁶A RNA modification co-transcriptionally. *Nature* 567, 414–419.
- Ip, J.Y., Schmidt, D., Pan, Q., Ramani, A.K., Fraser, A.G., Odom, D.T., and Blencowe, B.J. (2011). Global impact of RNA polymerase II elongation inhibition on alternative splicing regulation. *Genome Res.* 21, 390–401.
- Jia, G., Fu, Y., Zhao, X., Dai, Q., Zheng, G., Yang, Y., Yi, C., Lindahl, T., Pan, T., Yang, Y.-G., and He, C. (2011). N6-methyladenosine in nuclear RNA is a major substrate of the obesity-associated FTO. *Nat. Chem. Biol.* 7, 885–887.
- Kanhoush, R., Beenders, B., Perrin, C., Moreau, J., Bellini, M., and Penrad-Mobayed, M. (2010). Novel domains in the hnRNP G/RBMX protein with distinct roles in RNA binding and targeting nascent transcripts. *Nucleus* 1, 109–122.
- Karolchik, D., Hinrichs, A.S., Furey, T.S., Roskin, K.M., Sugnet, C.W., Haussler, D., and Kent, W.J. (2004). The UCSC Table Browser data retrieval tool. *Nucleic Acids Res.* 32, D493–D496.
- Kasowitz, S.D., Ma, J., Anderson, S.J., Leu, N.A., Xu, Y., Gregory, B.D., Schultz, R.M., and Wang, P.J. (2018). Nuclear m6A reader YTHDC1 regulates alternative polyadenylation and splicing during mouse oocyte development. *PLoS Genet.* 14, e1007412.
- Ke, S., Pandya-Jones, A., Saito, Y., Fak, J.J., Vågbo, C.B., Geula, S., Hanna, J.H., Black, D.L., Darnell, J.E., Jr., and Darnell, R.B. (2017). m⁶A mRNA modifications are deposited in nascent pre-mRNA and are not required for splicing but do specify cytoplasmic turnover. *Genes Dev.* 31, 990–1006.
- Kent, W.J., Zweig, A.S., Barber, G., Hinrichs, A.S., and Karolchik, D. (2010). BigWig and BigBed: enabling browsing of large distributed datasets. *Bioinformatics* 26, 2204–2207.
- Knuckles, P., Carl, S.H., Musheev, M., Niehrs, C., Wenger, A., and Bühler, M. (2017). RNA fate determination through cotranscriptional adenosine methylation and microprocessor binding. *Nat. Struct. Mol. Biol.* 24, 561–569.
- Kornblihtt, A.R., Schor, I.E., Alló, M., Dujardin, G., Petrillo, E., and Muñoz, M.J. (2013). Alternative splicing: a pivotal step between eukaryotic transcription and translation. *Nat. Rev. Mol. Cell Biol.* 14, 153–165.
- Kwon, I., Kato, M., Xiang, S., Wu, L., Theodoropoulos, P., Mirzaei, H., Han, T., Xie, S., Corden, J.L., and McKnight, S.L. (2013). Phosphorylation-regulated binding of RNA polymerase II to fibrous polymers of low-complexity domains. *Cell* 155, 1049–1060.
- Langmead, B., and Salzberg, S.L. (2012). Fast gapped-read alignment with Bowtie 2. *Nat. Methods* 9, 357–359.
- Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., Marth, G., Abecasis, G., and Durbin, R.; 1000 Genome Project Data Processing Subgroup (2009). The Sequence Alignment/Map format and SAMtools. *Bioinformatics* 25, 2078–2079.
- Liao, Y., Smyth, G.K., and Shi, W. (2014). featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* 30, 923–930.
- Lindorff-Larsen, K., Piana, S., Dror, R.O., and Shaw, D.E. (2011). How fast-folding proteins fold. *Science* 334, 517–520.
- Liu, J., Yue, Y., Han, D., Wang, X., Fu, Y., Zhang, L., Jia, G., Yu, M., Lu, Z., Deng, X., et al. (2014). A METTL3-METTL14 complex mediates mammalian nuclear RNA N6-adenosine methylation. *Nat. Chem. Biol.* 10, 93–95.
- Liu, N., Dai, Q., Zheng, G., He, C., Parisien, M., and Pan, T. (2015). N(6)-methyladenosine-dependent RNA structural switches regulate RNA-protein interactions. *Nature* 518, 560–564.
- Liu, N., Zhou, K.I., Parisien, M., Dai, Q., Diatchenko, L., and Pan, T. (2017). N6-methyladenosine alters RNA structure to regulate binding of a low-complexity protein. *Nucleic Acids Res.* 45, 6051–6063.
- Ljungman, M., and Hanawalt, P.C. (1996). The anti-cancer drug camptothecin inhibits elongation but stimulates initiation of RNA polymerase II transcription. *Carcinogenesis* 17, 31–35.
- Louloupi, A., Ntini, E., Conrad, T., and Ørom, U.A.V. (2018). Transient N6-Methyladenosine Transcriptome Sequencing Reveals a Regulatory Role of m6A in Splicing Efficiency. *Cell Rep.* 23, 3429–3437.
- Love, M.I., Huber, W., and Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* 15, 550.
- Mayer, A., di Iulio, J., Maleri, S., Eser, U., Vierstra, J., Reynolds, A., Sandstrom, R., Stamatoyannopoulos, J.A., and Churchman, L.S. (2015). Native elongating transcript sequencing reveals human transcriptional activity at nucleotide resolution. *Cell* 161, 541–554.
- Mayer, A., Landry, H.M., and Churchman, L.S. (2017). Pause & go: from the discovery of RNA polymerase pausing to its functional implications. *Curr. Opin. Cell Biol.* 46, 72–80.
- Meyer, K.D., Saletore, Y., Zumbo, P., Elemento, O., Mason, C.E., and Jaffrey, S.R. (2012). Comprehensive analysis of mRNA methylation reveals enrichment in 3' UTRs and near stop codons. *Cell* 149, 1635–1646.
- Mi, H., Huang, X., Muruganujan, A., Tang, H., Mills, C., Kang, D., and Thomas, P.D. (2017). PANTHER version 11: expanded annotation data from Gene Ontology and Reactome pathways, and data analysis tool enhancements. *Nucleic Acids Res.* 45 (D1), D183–D189.
- Moursy, A., Allain, F.H.-T., and Cléry, A. (2014). Characterization of the RNA recognition mode of hnRNP G extends its role in SMN2 splicing regulation. *Nucleic Acids Res.* 42, 6659–6672.

- Muñoz, M.J., de la Mata, M., and Kornblihtt, A.R. (2010). The carboxy terminal domain of RNA polymerase II and alternative splicing. *Trends Biochem. Sci.* **35**, 497–504.
- Nojima, T., Gomes, T., Grosso, A.R.F., Kimura, H., Dye, M.J., Dhir, S., Carmo-Fonseca, M., and Proudfoot, N.J. (2015). Mammalian NET-Seq Reveals Genome-wide Nascent Transcription Coupled to RNA Processing. *Cell* **161**, 526–540.
- Nojima, T., Rebelo, K., Gomes, T., Grosso, A.R., Proudfoot, N.J., and Carmo-Fonseca, M. (2018). RNA Polymerase II Phosphorylated on CTD Serine 5 Interacts with the Spliceosome during Co-transcriptional Splicing. *Mol. Cell* **72**, 369–379.
- O'Leary, N.A., Wright, M.W., Brister, J.R., Ciufo, S., Haddad, D., McVeigh, R., Rajput, B., Robbertse, B., Smith-White, B., Ako-Adjei, D., et al. (2016). Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. *Nucleic Acids Res.* **44** (D1), D733–D745.
- Okada, M., and Fukagawa, T. (2006). Purification of a protein complex that associates with chromatin. *PROTOCOL*. Published online December 15, 2006. <https://doi.org/10.1038/nprot.2006.417>.
- Ozdilek, B.A., Thompson, V.F., Ahmed, N.S., White, C.I., Batey, R.T., and Schwartz, J.C. (2017). Intrinsically disordered RGG/RG domains mediate degenerate specificity in RNA binding. *Nucleic Acids Res.* **45**, 7984–7996.
- Patil, D.P., Chen, C.-K., Pickering, B.F., Chow, A., Jackson, C., Guttman, M., and Jaffrey, S.R. (2016). m(6)A RNA methylation promotes XIST-mediated transcriptional repression. *Nature* **537**, 369–373.
- Ping, X.-L., Sun, B.-F., Wang, L., Xiao, W., Yang, X., Wang, W.-J., Adhikari, S., Shi, Y., Lv, Y., Chen, Y.-S., et al. (2014). Mammalian WTAP is a regulatory subunit of the RNA N6-methyladenosine methyltransferase. *Cell Res.* **24**, 177–189.
- Qiao, Y., Zhou, B., Zhang, M., Liu, W., Han, Z., Song, C., Yu, W., Yang, Q., Wang, R., Wang, S., et al. (2016). A Novel Inhibitor of the Obesity-Related Protein FTO. *Biochemistry* **55**, 1516–1522.
- Quinlan, A.R., and Hall, I.M. (2010). BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841–842.
- Ramírez, F., Dündar, F., Diehl, S., Grüning, B.A., and Manke, T. (2014). deepTools: a flexible platform for exploring deep-sequencing data. *Nucleic Acids Res.* **42**, W187–91.
- Roignant, J.-Y., and Soller, M. (2017). m⁶A in mRNA: An Ancient Mechanism for Fine-Tuning Gene Expression. *Trends Genet.* **33**, 380–390.
- Roundtree, I.A., Evans, M.E., Pan, T., and He, C. (2017). Dynamic RNA Modifications in Gene Expression Regulation. *Cell* **169**, 1187–1200.
- Schwartz, J.C., Wang, X., Podell, E.R., and Cech, T.R. (2013). RNA seeds higher-order assembly of FUS protein. *Cell Rep.* **5**, 918–925.
- Shashi, V., Xie, P., Schoch, K., Goldstein, D.B., Howard, T.D., Berry, M.N., Schwartz, C.E., Cronin, K., Sliwa, S., Allen, A., and Need, A.C. (2015). The RBMX gene as a candidate for the Shashi X-linked intellectual disability syndrome. *Clin. Genet.* **88**, 386–390.
- Shukla, S., Kavak, E., Gregory, M., Imashimizu, M., Shutinoski, B., Kashlev, M., Oberdoerffer, P., Sandberg, R., and Oberdoerffer, S. (2011). CTCF-promoted RNA polymerase II pausing links DNA methylation to splicing. *Nature* **479**, 74–79.
- Skinner, J.J., Yu, W., Gichana, E.K., Baxa, M.C., Hinshaw, J.R., Freed, K.F., and Sosnick, T.R. (2014). Benchmarking all-atom simulations using hydrogen exchange. *Proc. Natl. Acad. Sci. USA* **111**, 15975–15980.
- Slobodin, B., Han, R., Calderone, V., Vrieling, J.A.F.O., Loayza-Puch, F., Elkon, R., and Agami, R. (2017). Transcription Impacts the Efficiency of mRNA Translation via Co-transcriptional N6-adenosine Methylation. *Cell* **169**, 326–337.
- Sordet, O., Larochelle, S., Nicolas, E., Stevens, E.V., Zhang, C., Shokat, K.M., Fisher, R.P., and Pommier, Y. (2008). Hyperphosphorylation of RNA polymerase II in response to topoisomerase I cleavage complexes and its association with transcription- and BRCA1-dependent degradation of topoisomerase I. *J. Mol. Biol.* **381**, 540–549.
- Tang, C., Klukovich, R., Peng, H., Wang, Z., Yu, T., Zhang, Y., Zheng, H., Klungland, A., and Yan, W. (2018). ALKBH5-dependent m6A demethylation controls splicing and stability of long 3'-UTR mRNAs in male germ cells. *Proc. Natl. Acad. Sci. USA* **115**, E325–E333.
- Thandapani, P., O'Connor, T.R., Bailey, T.L., and Richard, S. (2013). Defining the RGG/RG motif. *Mol. Cell* **50**, 613–623.
- The Gene Ontology Consortium (2017). Expansion of the Gene Ontology knowledgebase and resources. *Nucleic Acids Res.* **45** (D1), D331–D338.
- Tsend-Ayush, E., O'Sullivan, L.A., Grützner, F.S., Onnebo, S.M.N., Lewis, R.S., Delbridge, M.L., Marshall Graves, J.A., and Ward, A.C. (2005). *RBMX* gene is essential for brain development in zebrafish. *Dev. Dyn.* **234**, 682–688.
- Wang, Y., Li, Y., Toth, J.I., Petroski, M.D., Zhang, Z., and Zhao, J.C. (2014). N6-methyladenosine modification destabilizes developmental regulators in embryonic stem cells. *Nat. Cell Biol.* **16**, 191–198.
- Xiang, Y., Laurent, B., Hsu, C.-H., Nachtergaele, S., Lu, Z., Sheng, W., Xu, C., Chen, H., Ouyang, J., Wang, S., et al. (2017). RNA m⁶A methylation regulates the ultraviolet-induced DNA damage response. *Nature* **543**, 573–576.
- Xiao, Y., Hsiao, T.-H., Suresh, U., Chen, H.-I.H., Wu, X., Wolf, S.E., and Chen, Y. (2014). A novel significance score for gene selection and ranking. *Bioinformatics* **30**, 801–807.
- Xiao, W., Adhikari, S., Dahal, U., Chen, Y.-S., Hao, Y.-J., Sun, B.-F., Sun, H.-Y., Li, A., Ping, X.-L., Lai, W.-Y., et al. (2016). Nuclear m(6)A Reader YTHDC1 Regulates mRNA Splicing. *Mol. Cell* **61**, 507–519.
- Yoon, K.-J., Ringeling, F.R., Vissers, C., Jacob, F., Pokrass, M., Jimenez-Cyrus, D., Su, Y., Kim, N.-S., Zhu, Y., Zheng, L., et al. (2017). Temporal Control of Mammalian Cortical Neurogenesis by m⁶A Methylation. *Cell* **171**, 877–889.
- Zhang, Y., Liu, T., Meyer, C.A., Eickhout, J., Johnson, D.S., Bernstein, B.E., Nussbaum, C., Myers, R.M., Brown, M., Li, W., and Liu, X.S. (2008). Model-based analysis of ChIP-Seq (MACS). *Genome Biol.* **9**, R137.
- Zhao, X., Yang, Y., Sun, B.-F., Shi, Y., Yang, X., Xiao, W., Hao, Y.-J., Ping, X.-L., Chen, Y.-S., Wang, W.-J., et al. (2014). FTO-dependent demethylation of N6-methyladenosine regulates mRNA splicing and is required for adipogenesis. *Cell Res.* **24**, 1403–1419.
- Zheng, G., Dahl, J.A., Niu, Y., Fedorcsak, P., Huang, C.-M., Li, C.-J., Vågbo, C.B., Shi, Y., Wang, W.-L., Song, S.-H., et al. (2013). ALKBH5 is a mammalian RNA demethylase that impacts RNA metabolism and mouse fertility. *Mol. Cell* **49**, 18–29.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit anti-hnRNPG antibody	Abcam	ab190352
Goat anti-rabbit IgG Alexa Fluor 647 antibody	Thermo	A-31573; RRID:AB_2536183
Rabbit anti-DDDDK antibody	Abcam	ab1162; RRID:AB_298215
Rabbit anti-hnRNPG antibody	Cell Signaling Technology	14794; RRID:AB_2798614
Rabbit anti-hnRNPG antibody	Novus Biologicals	NBP2-34152
Rabbit isotype control antibody	Abcam	ab199376
Rabbit anti-RNAPII S5P antibody	Abcam	ab5131; RRID:AB_449369
Mouse anti-RNAPII S2P H5 antibody	BioLegend	920204; RRID:AB_2616695
Rabbit anti-METTL3 antibody	Proteintech	15073-1-AP; RRID:AB_2142033
Mouse anti-GST antibody	Cell Signaling Technology	2624; RRID:AB_2189875
Goat anti-GAPDH antibody-HRP	Genscript	A00192-40
Mouse anti-rabbit light chain antibody-HRP	Abcam	ab99697; RRID:AB_10673897
Goat anti-rabbit IgG antibody-HRP	Abcam	ab97051; RRID:AB_10679369
Rat anti-mouse light chain antibody-HRP	Abcam	ab99632; RRID:AB_10697414
Goat anti-mouse IgG antibody-HRP	Abcam	ab97023; RRID:AB_10679675
Mouse anti-RNA Polymerase II RPB1 Antibody	BioLegend	664912; RRID:AB_2650945
Mouse Anti-RNA Polymerase II CTD (Ser2)	MBL INTERNATIONAL	MAB10602; RRID:AB_2747403
ChIP-seq Spike-in Antibody	Active motif	61686; RRID:AB_2737370
Rabbit Anti-SNRP70/U1-70K antibody	Abcam	ab83306; RRID:AB_10673827
Rabbit Histone H3 (tri methyl K4) antibody	Abcam	ab8580; RRID:AB_306649
Chemicals, Peptides, and Recombinant Proteins		
α -amanitin	Sigma	A2263
Actinomycin D	Sigma	A9415
5,6-dichloro-1- β -D-ribofuranosylbenzimidazole (DRB)	Sigma	D1916
Camptothecin	Sigma	C9911
GST-tagged human RNAPII CTD (GST-CTD)	Sigma	SRP2120
CDK7-Cyclin H-MNAT1	Thermo	PV3868
CDK9-Cyclin T1	Sigma	14-685
4-Thiouridine, $\geq 98\%$	Sigma	T4509
ChIP-seq Spike-in Chromatin	Active motif	53083
Deposited Data		
Raw and analyzed mRNA-seq data	This paper	GEO: GSE114311
Mendeley Dataset	This paper	https://doi.org/10.17632/5hrrjww3xc.1
Experimental Models: Cell Lines		
HEK293T/17 cell line	ATCC	CRL11268
Oligonucleotides		
See “Oligonucleotides” section under “ Method Details ”		N/A
Recombinant DNA		
pCMV3-Flag negative control vector	Sino Biological	CV016
pCMV3-Flag-RBMX plasmid	Sino Biological	HG16560-NF
Software and Algorithms		
featureCounts	Liao et al., 2014	http://subread.sourceforge.net
DEseq2	Love et al., 2014	https://bioconductor.org/biocLite.R

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Gene ontology enrichment analysis tool	Ashburner et al., 2000; Mi et al., 2017; The Gene Ontology Consortium, 2017	https://geneontology.github.io/page/go-enrichment-analysis/
BEDTools intersect function version 2.17.0	Quinlan and Hall, 2010	https://github.com/arq5x/bedtools2
Bedtools 2.28.0	Quinlan and Hall, 2010	https://bedtools.readthedocs.io/en/latest/
ImageJ	Open source	v1.50i
DYNAMICS	Wyatt	v7.1.7
SlideBook	Intelligent Imaging Innovations	v6.0.15
R statistical software	Open source	v3.5.0
Bowtie2	Langmead and Salzberg, 2012	http://bowtie-bio.sourceforge.net/bowtie2/manual.shtml
samtools	Li et al., 2009	http://samtools.sourceforge.net/
Macs2	Zhang et al., 2008	https://github.com/taoliu/MACS
deeptools	Ramírez et al., 2014	https://deeptools.readthedocs.io/en/develop/
bedGraphToBigWig	Kent et al., 2010	http://hgdownload.cse.ucsc.edu/admin/exe/

LEAD CONTACT AND MATERIALS AVAILABILITY

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Tao Pan (taopan@uchicago.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Human embryonic kidney (HEK) cell line HEK293T/17 (CRL11268) was obtained from the American Type Culture Collection (ATCC) and cultured in Dulbecco's Modified Eagle's Medium (DMEM) with high glucose and L-glutamine, without sodium pyruvate (HyClone, SH30022.01) in a 37°C incubator with a humidified atmosphere of 5% CO₂. For Figure 1B, HEK293T cells were treated with 0.5% v/v dimethyl sulfoxide (DMSO), 5 µg/mL actinomycin D (A9415, Sigma), or 100 µM 5,6-dichloro-1-β-D-ribofuranosylbenzimidazole (DRB) (D1916, Sigma) for 2 hours before collection of cells for chromatin extraction. For Figure 1C, HEK293T cells were treated with 0.12% v/v DMSO or 6 µM camptothecin (C9911, Sigma) for 5 hours before collection of cells for chromatin extraction. For Figure S1C, HEK293T cells were treated with +2% v/v H₂O or 20 µg/mL α-amanitin (A2263, Sigma) for 9 hours, or with 5 µg/mL actinomycin D (A9415, Sigma) for 2 hours, before fixation for immunofluorescence. For Figures 1E and S3A, 24 ng/mL of either pCMV3-Flag negative control vector (CV016, Sino Biological) or pCMV3-Flag-RBMX plasmid (WT, RRMmut, RGG1mut, or RGG2mut), as well as 10 nM hnRNPG siRNA (SI00700077, QIAGEN), were transfected into HEK293T cells using Lipofectamine 2000 (11668019, Thermo), and the cells were collected for chromatin or RNA extraction 68 hours after transfection. For Figure 1F, 20 nM control siRNA (1027281, QIAGEN) or METTL3 siRNA (SI04317096, QIAGEN) was transfected into HEK293T cells using Lipofectamine RNAiMAX (13778150, Thermo), and the cells were collected for chromatin extraction 60 hours after transfection. For Figure S1A, 20 nM control siRNA (1027281, QIAGEN) or hnRNPG siRNA (G1: SI00700077, QIAGEN; G2: SI00700084, QIAGEN) was transfected into HEK293T cells using Lipofectamine RNAiMAX, and the cells were collected for chromatin extraction 62 hours after transfection.

METHOD DETAILS**Cloning and purification of hnRNPG**

Repeated attempts by us to overexpress and purify full-length hnRNPG protein in *E. coli* using multiple different plasmid constructs and expression strategies were unsuccessful, likely due to protease cleavage of the low-complexity hnRNPG protein in cells. We therefore resorted to overexpress the full-length protein from baculovirus expression system where the full-length protein was secreted to avoid proteolysis (Araç et al., 2012).

The sequence encoding full-length human hnRNPG protein was amplified from human HeLa cDNA libraries (637203, Clontech) and subcloned into the pGEX-6p-1 vector using BamHI and XhoI restriction sites. Plasmids encoding the hnRNPG mutants RRMmut, RGG1mut, and RGG2mut were prepared by QuikChange mutagenesis (200524, Agilent) and Gibson assembly (E2611L, New England BioLabs), with the following mutations in the encoded proteins: K9A, F11A, R49A, F51A, and F53A in RRMmut; R110A, R113A, R120A, and R125A in RGG1mut; R369A, R373A, and R377A in RGG2mut. The mutant hnRNPG sequences were cloned into pCMV3-Flag-RBMX (HG16560-NF, Sino Biological) by Gibson assembly for expression in human cells. The wild-type and mutant hnRNPG sequences were cloned into the pAcGP67a vector using BamHI and NotI restriction sites for expression in insect cells.

A His₈ tag was added to the N terminus for affinity purification. A fast-folding variant of protein G, NuG2b (DTYKLVIVLNGTTFTYTT EAVDAATAEKVKQYANDAGVDGEWYDAATKFTVTET) (Lindorff-Larsen et al., 2011; Skinner et al., 2014), was added to the N terminus to increase protein stability.

A baculovirus expression system was used for expression of proteins in High Five insect cells as previously described (Langmead and Salzberg, 2012). The secreted proteins were purified using nickel nitrilotriacetic agarose (Ni-NTA) resin (30230, QIAGEN). The resin was washed with 10 mM Tris-Cl (pH 7.4), 1 M NaCl, 2.5 mM MgCl₂, 10% v/v glycerol buffer. The hnRNPG protein was released from the resin by cleavage C-terminal to the His₈-NuG2b tag with His-tagged Pre-Scission Protease (Z03092, GenScript) in storage buffer (10 mM Tris-Cl (pH 7.4), 500 mM NaCl, 2.5 mM MgCl₂, 10% v/v glycerol) at 4°C overnight, and the protein was concentrated with a 30 kDa centrifugal filter (UFC803024) and either stored at 4°C or flash-frozen in liquid nitrogen and stored at –80°C. Immediately before use, hnRNPG protein stocks were spun at 21 K × g at 4°C for 10 minutes, and the supernatant was used as the new hnRNPG stock. The concentrations of hnRNPG stocks were measured by Bradford assay (23236, Thermo).

Immunofluorescence

HEK293T cells were grown in tissue-culture-treated 8-well slides (80826, ibidi), fixed with 4% w/v formaldehyde, permeabilized with phosphate-buffered saline (PBS) (10 × solution: SH30258.01, HyClone) containing 0.2% v/v Triton X-100, and blocked overnight at 4°C in blocking buffer (PBS with 2% w/v bovine serum albumin (A7030, Sigma)). After blocking, the cells were incubated with 365 ng/mL rabbit anti-hnRNPG antibody (ab190352, Abcam) in blocking buffer for 1 hour at room temperature, incubated with 1 μg/mL goat anti-rabbit IgG Alexa Fluor 647 antibody (A-31573, Thermo) in blocking buffer for 1 hour, stained with PBS containing 0.1 μg/mL 4',6-diamidino-2-phenylindole (DAPI) (D1306, Thermo) for 2 minutes, and covered with ibidi mounting medium (50001, ibidi). Between steps, the cells were washed 2–3 times with PBS for 5 minutes per wash. The slides were imaged on an Olympus DSU spinning disk confocal microscope at the University of Chicago Integrated Light Microscopy Core Facility.

Preparation of cell extracts

HEK293T cells were washed and detached from the cell culture plate with PBS, and then pelleted at 500 × g for 3 minutes. For extraction of whole cell lysate, the cells were resuspended in whole cell lysis buffer: 300 mM NaCl, 100 mM Tris-Cl (pH 8), 0.2 mM ethylenediaminetetraacetic acid (EDTA), 0.1% v/v Triton X-100, and 10% v/v glycerol supplemented with freshly added 1% v/v protease inhibitor (25955-11, Nacalai USA) and 1% v/v phosphatase inhibitor (07575-51, Nacalai USA). After rotating at 4°C for 30 minutes, cell debris was pelleted at 16 K × g at 4°C for 5 minutes, and the supernatant whole cell lysate was collected and stored at –20°C.

Cell fractionation and chromatin extraction were performed based on published protocols (Ferrer et al., 2018; Groisman et al., 2003; Okada and Fukagawa, 2006). HEK293T cells were resuspended in cytoplasmic extraction buffer: 10 mM Tris-Cl (pH 7.4), 10 mM KCl, and 0.1% v/v Triton X-100 supplemented with freshly added 1% v/v protease and phosphatase inhibitors. After incubating on ice for 20 minutes, nuclei were pelleted at 12 K × g at 4°C for 10 minutes, and the supernatant was collected as the cytoplasmic extract. The pellet was washed once with cytoplasmic extraction buffer. The nuclei were resuspended with nuclear extraction buffer: 10 mM Tris-Cl (pH 7.4), 0.2 mM MgCl₂, and 1% v/v Triton X-100 supplemented with freshly added 1% v/v protease and phosphatase inhibitors. After incubating on ice for 15 minutes, chromatin was pelleted at 12 K × g at 4°C for 10 minutes, and the supernatant was collected as the nucleoplasmic extract. The chromatin pellets were resuspended in 5 U/mL micrococcal nuclease (N3755, Sigma), 20 mM Tris-Cl (pH 7.4), 100 mM KCl, 2 mM MgCl₂, 1–3 mM CaCl₂, 0.3 M sucrose, and 0.1% v/v Triton X-100 supplemented with freshly added 1% v/v protease and phosphatase inhibitors. After rotating at 4°C for 1 hour, the digestion reaction was stopped by adding 5 mM EDTA. After centrifuging at 2 K × g at 4°C for 5 minutes, the supernatant was collected as the chromatin extract.

Immunoprecipitation

For immunoprecipitation, HEK293T whole cell lysate or chromatin extract was combined with 28 μg/mL rabbit anti-DDDDK antibody (ab1162, Abcam), 20 μg/mL rabbit anti-hnRNPG antibody (Ab #1 or not specified: ab190352, Abcam), 1/20 v/v rabbit anti-hnRNPG antibody (Ab #2: 14794, Cell Signaling Technology; Ab #3: NBP2-34152, Novus Biologicals), or 20 μg/mL rabbit isotype control antibody (ab199376, Abcam) in a total volume of 90–100 μL per 100-mm plate of cells. After rotating at 4°C overnight, 1.5 mg of protein A or protein G Dynabeads (10002D or 10004D, Thermo) were added. After rotating at 4°C for 2 hours, the tubes were placed on a magnetic separation rack, and the supernatant unbound fraction was collected and stored at –20°C. The beads were washed four times with 200 μL of wash buffer (300 mM NaCl, 100 mM Tris-Cl (pH 8), 0.2 mM EDTA, and 0.1% v/v Triton X-100). Finally, 30 μL of 4 × LDS sample buffer (NP0008, Thermo) were added, and the tubes were incubated at 95°C for 5 minutes and then placed on a magnetic rack. The supernatant immunoprecipitate (IP) was collected and stored at –20°C. The concentrations of the input and unbound fractions were measured by Bradford assay. For western blotting, 5 μg of input (whole cell lysate or chromatin extract) or unbound fraction was combined with 1 × LDS and 250 mM dithiothreitol (DTT), and 5 μL (one sixth) of the IP fraction was combined with 250 mM DTT.

GST-CTD pull-down

To phosphorylate the GST-CTD, 500 ng of recombinant GST-tagged human RNAPII CTD (SRP2120, Sigma) were combined with 240 ng of CDK7–Cyclin H–MNAT1 (PV3868, Thermo) or CDK9–Cyclin T1 (14-685, Sigma) in 20 μL of 8 mM 3-(N-morpholino)propanesulfonic

acid (MOPS) (pH 7), 0.2 mM EDTA, and 1 mM MgCl₂ supplemented with freshly added 0.1 mM ATP and 0.25 mM DTT, and then incubated at 30°C for 1 hour. Control reactions without GST-CTD, without kinase, or without ATP were incubated under the same conditions. For the GST-CTD pull-down and immunoprecipitation assays, this 20 μ L reaction was combined with 60 μ L of wash buffer and 100 pmol of hnRNPG in 20 μ L of storage buffer (10 mM Tris-Cl (pH 7.4), 500 mM NaCl, 2.5 mM MgCl₂, 10% v/v glycerol), and then rotated at 4°C overnight in a total volume of 100 μ L. One tenth (10 μ L) of the binding mixture was taken as the input. A 20 μ L volume of glutathione magnetic agarose beads (78601, Thermo) for the pull-down (Figures 2A and 2C), or GST mouse monoclonal antibody magnetic bead conjugate (11847, Cell Signaling Technology) for the immunoprecipitation (Figure 2B), was blocked in 100 μ L of 1 mg/mL BSA in wash buffer at 4°C for 1 hour, resuspended in 20 μ L of wash buffer, and added to the remaining 90 μ L of the binding mixture. After rotating at 4°C for 2 hours, the tubes were placed on a magnetic rack, and the supernatant unbound fraction was collected. The beads were washed four times with 200 μ L of wash buffer. For elution, the beads were resuspended in 30 μ L of 50 mM reduced glutathione in wash buffer and incubated at 22°C, 300 rpm for 20 minutes for the pull-down, or resuspended in 30 μ L of 4 $\times$ LDS and boiled at 95°C for 5 minutes for the immunoprecipitation. The tubes were placed on a magnetic rack, and the supernatant eluate was collected and stored at –20°C. For blotting, 10 μ L of the input fraction (one tenth of total input) or 10 μ L of the eluate (one third) were prepared in solutions with final concentrations of 1–2 $\times$ LDS and 250 mM DTT. The pull-down samples were visualized by western blotting for hnRNPG (below), while the immunoprecipitation samples were visualized by Sypro Ruby protein blot staining (S-11791, Thermo) according to the manufacturer's instructions. Both the pull-down and immunoprecipitation samples were also visualized by western blotting for GST (below).

Western blotting

All samples were incubated at 95°C for 10 minutes, separated on a 4%–12% polyacrylamide Bis-Tris protein gel (NP0336BOX, Thermo), and transferred to polyvinylidene fluoride membranes (IPVH00010, Millipore). The membranes were blocked in 5% w/v milk (170-6404, Bio-Rad), 20 mM Tris-Cl (pH 7.6), 150 mM NaCl, and 0.1% v/v Triton X-100. The blots were probed with 0.7 μ g/mL anti-RNAPII S5P antibody (ab5131, Abcam), 1 μ g/mL anti-RNAPII S2P H5 antibody (920204, BioLegend), 0.365 μ g/mL anti-hnRNPG antibody (ab190352, Abcam), 1 μ g/mL anti-DDDDK (anti-FLAG) antibody (ab1162, Abcam), 0.23 μ g/mL anti-METTL3 antibody (15073-1-AP, Proteintech), 1/1000 v/v anti-GST antibody (2624, Cell Signaling Technology), 1/3000 v/v anti-SNRP70 antibody (ab83306, Abcam), 1/5000 v/v anti-H3K4me3 (ab8580, Abcam), or 0.05 μ g/mL anti-GAPDH antibody (A00192-40, GenScript), followed by 0.5 μ g/mL mouse anti-rabbit light chain (ab99697, Abcam), 0.03 μ g/mL goat anti-rabbit IgG (ab97051, Abcam), 1/2000 v/v rat anti-mouse light chain (ab99632, Abcam), or 0.05 μ g/mL goat anti-mouse IgG antibody conjugated to horseradish peroxidase (ab97023, Abcam).

Cell proliferation

After transfecting HEK293T cells with WT or RGG2mut plasmid (24 ng/ml final) and RBMX siRNA (10 nM final) for 68 hours as described above, cells were counted (Countess II Automated Cell Counter, Thermo Fisher) and seeded onto 96-well plates (3000 cells per well, n = 8 for each cell type and time point). Cell proliferation was measured using MTT colorimetric assay (CytoSelect MTT Cell Proliferation Assay, Cell Biolabs Inc) at 0, 48, 72, and 96 hours according to the manufacturer's instructions. After incubation with MTT reagent (3.5 hours, 37°C, 5% CO₂), the Detergent Solution was added to each well to dissolve formazan crystals (16 hours, room temperature). The absorbance at 560 nm was measured using a Synergy Neo Plate Reader (BioTek). At each time point, the absorbance of the media was subtracted from each measurement.

Oligonucleotides

The following RNA oligonucleotides were synthesized and purified by high-performance liquid chromatography and/or denaturing polyacrylamide gel electrophoresis, as previously described (Dai et al., 2007; Liu et al., 2017).

REAGENT or RESOURCE	SOURCE	IDENTIFIER
2515-A RNA: 5'- AAUGUGAAGGACUUUCGUAACGGAAGUAAUUCAA	Liu et al., 2017	N/A
2515-m ⁶ A RNA: 5'- AAUGUGAAGGm ⁶ ACUUUCGUAACGGAAGUAAUUCAA	Liu et al., 2017	N/A
2515-A-Biotin RNA: 5'- AAUGUGAAGGACUUUCGUAACGGAAGUAAUUCAA-Biotin	Liu et al., 2017	N/A
2515-m ⁶ A-Biotin RNA: 5'- AAUGUGAAGGm ⁶ ACUUUCGUAACGGAAGUAAUUCAA-Biotin	Liu et al., 2017	N/A

The following DNA oligonucleotides were ordered from Integrated DNA Technologies, purified by denaturing polyacrylamide gel electrophoresis, annealed in 50 mM Tris-Cl (pH 7.5), 500 mM KCl at 94°C for 1 minute, and incubated at room temperature for 3–5 minutes. The resulting DNA templates were used for *in vitro* transcription of the SMNexon7 and WEc RNAs with T7 RNA polymerase (M0251, New England BioLabs), and the RNA transcripts were precipitated and purified by denaturing gel electrophoresis.

REAGENT or RESOURCE	SOURCE	IDENTIFIER
SMNexon7-Fw DNA: 5'-TAATACGACTCACTATAGGTTTGTAGACAAAATCAAAAAGAAGGAAGGTGCTC ACATTCCTTAAATTAAGGA	Integrated DNA Technologies	N/A
SMNexon7-Rv DNA: 5'-TCCTTAATTTAAGGAATGTGAGCACCTTCTTCTTTTGTCTTAAACC TATAGTGAGTCGTATTA	Integrated DNA Technologies	N/A
WEc-Fw DNA: 5'-TAATACGACTCACTATAGGGTACGACGGATATCGTGGGGGGGAAATTGCTTTTCG GTTCCGACTCTG	Integrated DNA Technologies	N/A
WEc-Rv DNA: 5'-CAGAGTCGGAACCGAAAGCAATTTCCCCCCCCACGATATCCGTCGTACCTATAG TGAGTCGTATTA	Integrated DNA Technologies	N/A

The following DNA oligonucleotides were ordered from Integrated DNA Technologies, purified by denaturing polyacrylamide gel electrophoresis, and used for RT-PCR validation of splicing changes and for RT-qPCR validation of transcript abundance changes detected by mRNA sequencing:

REAGENT or RESOURCE	SOURCE	IDENTIFIER
BRCA1_Fw DNA: 5'-GTGGTCAATGGAAGAAACCACC	Integrated DNA Technologies	N/A
BRCA1_Rv DNA: 5'-CCTTCACACAGAAGCACAC	Integrated DNA Technologies	N/A
DNAH10_Fw DNA: 5'-AACAGGACTCTCCCCACTT	Integrated DNA Technologies	N/A
DNAH10_Rv DNA: 5'-GAGACTTCGTCATCCGGACC	Integrated DNA Technologies	N/A
TNN_Fw DNA: 5'-AATTGCGTCCATAGAGCTGC	Integrated DNA Technologies	N/A
TNN_Rv DNA: 5'-GGTGGGCATGAATGGGAGTT	Integrated DNA Technologies	N/A
TAF7L_Fw DNA: 5'-TGACGCTTTTGTCCCTTCAG	Integrated DNA Technologies	N/A
TAF7L_Rv DNA: 5'-CATGGTCGAGTGGGGTCTTT	Integrated DNA Technologies	N/A
GYS2_Fw DNA: 5'-GCATGAGGAGGAGCACTCATT	Integrated DNA Technologies	N/A
GYS2_Rv DNA: 5'-GTGGGGTAGAGAAGCTGCAT	Integrated DNA Technologies	N/A
LINC00926_Fw DNA: 5'-GGTATCTTGCCCTTCTGGG	Integrated DNA Technologies	N/A
LINC00926_Rv DNA: 5'-CCCACCTCTGAAAAGCCCAT	Integrated DNA Technologies	N/A
CYP2E1_Fw DNA: 5'-AATTGACAGGGTGATTGGGC	Integrated DNA Technologies	N/A
CYP2E1_Rv DNA: 5'-TCATGCACACAGCATCCAT	Integrated DNA Technologies	N/A
KRTAP19-1_Fw DNA: 5'-GGAGGATACGGATTCTCTGGC	Integrated DNA Technologies	N/A
KRTAP19-1_Rv DNA: 5'-TGGAATGAAAGCACGGGACA	Integrated DNA Technologies	N/A
ACTB_Fw DNA: 5'-AGGACTTTGATTGCACATTGTT	Integrated DNA Technologies	N/A
ACTB_Rv DNA: 5'-TGGGGTGGCTTTTAGGATGG	Integrated DNA Technologies	N/A

Surface plasmon resonance

Surface plasmon resonance data were collected on a BioRad ProteOn XPR36 instrument at the University of Chicago BioPhysics Core Facility. The RNA oligonucleotides 2515-A-Biotin and 2515-m⁶A-Biotin were immobilized on separate lanes of an NLC sensor chip (1765021 Bio-Rad). Recombinant hnRNPG protein was flowed over the chip at a concentration of 3 μ M with 0–12 μ M S5P peptide (ab18488, Abcam) in filtered and degassed 10 mM 4-(2-hydroxyethyl)piperazine-1-ethanesulfonic acid (HEPES) (pH 8.6), 500 mM NaCl, 3 mM EDTA, and 0.05% v/v Polysorbate 20 buffer.

Limited proteolysis

The RNA was diluted to a concentration of 25 μ M in 30 mM Tris-Cl (pH 7.4) and incubated at 90°C for 1 minute, and then at room temperature for 3–5 minutes. For limited proteolysis, 200 pmol of wild-type or mutant hnRNPG were combined 200 pmol of RNA in 14 mM Tris-Cl (pH 7.4), 400 mM NaCl, 2 mM MgCl₂, and 8% v/v glycerol in a total volume of 40 μ L, and the mixture was incubated for 10 minutes at room temperature, and then placed on ice. Proteinase K (EO0491, Thermo) was added for a ratio of 1:700 w/w proteinase:hnRNPG, and the reaction was incubated on ice. Time points were taken at 5, 15, 30, and 60 minutes by combining 9.21 μ L of the reaction with stop solution for a final concentration of 1 $\times$ LDS, 1% v/v protease inhibitor, and incubating at 95°C for 5 minutes before placing on dry ice. The 0-minute time point was prepared by preparing 1.7 μ g of hnRNPG in 1 $\times$ LDS, 1% v/v protease inhibitor. Immediately after the 60-minute time point was collected, all of the samples were incubated at 95°C for 5 minutes, separated on a 4%–12% polyacrylamide BiEES-Tris gel, and stained with PageBlue Protein Staining Solution (24620, Thermo) according to the manufacturer's instructions.

Dynamic light scattering

For Figure S2C, 9 μ M wild-type or mutant hnRNP G protein was prepared in 14 mM Tris-Cl (pH 7.4), 400 mM NaCl, 2 mM $MgCl_2$, and 8% v/v glycerol, with or without 7.2 μ M RNA, for a total volume of 40 μ L. For Figure S2E, 9 μ M wild-type hnRNP G protein was prepared in storage buffer (10 mM Tris-Cl (pH 7.4), 500 mM NaCl, 2.5 mM $MgCl_2$, 10% v/v glycerol) with or without 36 μ M S5P or S2P peptide (ab18488 and ab12793, Abcam), for a total volume of 40 μ L. The mixture was incubated for 10 minutes at room temperature, and then placed on ice. Light scattering measurements were collected at 4°C on a Wyatt DynaPro NanoStar instrument and analyzed using the DYNAMICS software at the University of Chicago BioPhysics Core Facility. Each measurement was the average of ten 20 s acquisitions. Each acquisition was fit to an auto-correlation curve using the regularization algorithm, with upper and lower cut-offs of 10^7 μ s and 1.5 μ s. Acquisitions with an auto-correlation curve baseline outside the 0.99–1.01 range or error sum-of-squares > 100 were filtered out before averaging.

HnRNP G spin-down assay for RNA binding

32 P-labeled RNA was pre-mixed $\pm$ hnRNP G protein at 0.5 M NaCl, then transferred to 0.15 M NaCl followed by centrifugation. Pellets were dissolved with 0.5 M NaCl buffer in the same volume as supernatant, spotted in μ L indicated, and followed by phosphorimaging.

20 pmol of MALAT1 hairpin containing A (2515-A) or m⁶A (2515-m⁶A) were 5' 32 P-labeled using T4 PNK and purified by denaturing gel electrophoresis and stored in water. 8 μ L of 32 P-RNA was mixed with 2 μ L 100 mM TrisHCl (pH 7.4) and renatured by heating at 85°C for 2 minutes, then placed at room temperature for 5 minutes. 10 μ L 2X Buffer A (40 mM Tris-Cl (pH 7.4), 1 M NaCl, 5 mM $MgCl_2$, 0.4 mM EDTA) was then added, and this mixture was split into 5x 3.5 μ L samples. 1.8 μ L 9.2 μ M hnRNP G protein in storage buffer (10 mM TrisHCl (pH 7.4), 500 mM NaCl, 2.5 mM $MgCl_2$, 10% v/v glycerol) and 4.7 μ L Buffer A (20 mM TrisHCl (pH 7.4), 500 mM NaCl, 2.5 mM $MgCl_2$, 0.2 mM EDTA) was then added. The final concentration for 32 P-RNA was less than 0.1 μ M. For samples with unlabeled RNA (2515-A or 2515-m⁶A), RNAPII CTD phosphor-S5 peptide (ab18488, abcam), or RNAPII CTD phosphor-S2 peptide (ab12793, abcam), they were diluted to the appropriate concentration in Buffer A and then added in lieu of Buffer A. Control reactions without hnRNP G substituted storage buffer (10 mM Tris-Cl (pH 7.4), 500 mM NaCl, 2.5 mM $MgCl_2$, 10% v/v glycerol) for hnRNP G. The binding mixtures were rotated at 4°C overnight in a total volume of 10 μ L.

HnRNP G protein precipitation was initiated by diluting the above binding mixture to a final NaCl concentration of 150 mM by adding 23.3 μ L Buffer B (20 mM TrisHCl (pH 7.4), 0 mM NaCl, 2.5 mM $MgCl_2$, 0.2 mM EDTA) and rotated at 4°C for 1 hour. The mixtures were then centrifuged for 30 minutes at 17,000 g and 4°C to pellet. Supernatant was separated from the pellet, and the pellet was resuspended in 33.3 μ L Buffer A. 1, 2.5, and 6.25 μ L of each sample was blotted onto filter paper. This blot was exposed on a phosphor-imager plate and imaged using a BioRad Personal Molecular Imager. The resulting image was quantified using the Quantity One software.

HnRNP G spin-down assay for CTD binding

HnRNP G protein $\pm$ GST-CTD $\pm$ phosphorylation by CDK7 or CDK9 were pre-mixed at 0.5 M NaCl, then transferred to 0.15 M or 0.25 M NaCl followed by centrifugation.

To phosphorylate the GST-CTD, 0.75 ng of recombinant GST-tagged human RNAPII CTD (SRP2120, Sigma) were combined with 0.36 ng of CDK7–Cyclin H–MNAT1 (PV3868, Thermo) or CDK9–Cyclin T1 (14-685, Sigma) in 25 μ L of 8 mM 3-(N-morpholino) propanesulfonic acid (MOPS) (pH 7), 0.2 mM EDTA, and 1 mM $MgCl_2$ supplemented with freshly added 0.1 mM ATP and 0.25 mM DTT, and then incubated at 30°C for 1 hour. Control reactions without ATP were incubated under the same condition. Binding mixtures using 2x GST-CTD instead utilized 1.5 ng of recombinant GST-tagged human RNAPII CTD and 0.72 ng of CDK7 or CDK9.

Initial binding mixtures were generated by adding 2.45 μ L of the above phosphorylation reaction to 1.1 μ L of either 9.2 or 18.4 μ M hnRNP G in storage buffer (10 mM TrisHCl (pH 7.4), 500 mM NaCl, 2.5 mM $MgCl_2$, 10% v/v glycerol) and 2.45 μ L of 2x Buffer A (40 mM TrisHCl (pH 7.4), 1 M NaCl, 5 mM $MgCl_2$, 0.4 mM EDTA). For binding mixtures with RNA, 1.63 μ L of 3X Buffer A (60 mM TrisHCl (pH 7.4), 1.5 M NaCl, 7.5 mM $MgCl_2$, 0.6 mM EDTA) and 0.82 μ L of 25 or 50 μ M 2515-A or 2515-m⁶A was added instead. The binding mixtures were rotated overnight at 4°C overnight in a total volume of 6 μ L.

Protein precipitation was initiated by diluting binding mixtures to either final 150 or 250 mM NaCl by adding either 14 μ L Buffer B (20 mM Tris-Cl (pH 7.4), 0 mM NaCl, 2.5 mM $MgCl_2$, 0.2 mM EDTA) or 2 μ L 2X Buffer A and 12 μ L Buffer B, respectively. These diluted binding mixtures were rotated for 1 hour at 4°C in a total volume of 20 μ L. The binding mixtures were then centrifuged for 1 hour at 17,000 g and 4°C to pellet. Supernatant was separated from the pellet, and the pellet was resuspended in 20 μ L Buffer B. 20 μ L of 2x LDS (NP0008, Thermo), 50 mM EDTA was then added to each sample, and they were then incubated at 95°C for 10 min. 20 μ L of each sample was then loaded onto 12-well 4%–12% polyacrylamide Bis-Tris gels (NP03322, Invitrogen) alongside a Novex Sharp Pre-stained Protein Standard (LC5800, Thermo) and ran at 150V for 1 hour. The gels were then silver stained using the Pierce Silver Stain Kit (24612, Thermo). Gels were imaged using a UVP BioDoc-It Imaging System. The intensities of each band were quantified using Image Studio software.

mRNA sequencing

HEK293T cells were transfected with hnRNP G siRNA and pCMV3-Flag (NCV) or pCMV3-Flag-RBMX (WT, RRMmut, RGG1mut, or RGG2mut) as described above, with three 100-mm-plate biological replicates for each of the five different plasmid transfections.

Total RNA was extracted from HEK293T cells by Trizol (15596026, Thermo) extraction followed by an additional chloroform extraction. Libraries were prepared using the Tru-Seq Stranded mRNA LT Sample Prep Kit (RS-122-9005DOC, Illumina). The libraries were sequenced twice, and each time the 15 libraries were sequenced in two lanes of one flow cell by Illumina HiSeq 4000 with paired-end 100-bp reads at the University of Chicago Genomics Facility. The reads from both sequencing runs were combined for the analysis.

Chromatin PAR-CLIP

Three plates of HEK293T cells per biological replicate were fed with 200 μ M 4-Thiouridine (T4905, Sigma) for 14 hours before collection. For each plate of cells, the media was aspirated, cells were washed twice with 5 mL ice-cold DPBS, and crosslinked twice on ice by 0.15 J/cm² 365 nm Ultraviolet light (CL-1000 Ultraviolet Crosslinker, UVP). The cells were then collected in ice-cold DPBS, collected by cell lifter, and pelleted at 500 $\times$ g for 5 minutes at 4°C.

Chromatin extraction was performed based on a published protocol (Brugiolo et al., 2017). Briefly, the cell pellet was flicked to resuspend in 1.8 mL ice-cold NP-40 lysis buffer (10 mM Tris-HCl [pH 7.5], 0.05% NP40, 150 mM NaCl, 1% protease and phosphatase inhibitor (78440, Thermo)) and incubate on ice for 5 min. To separate the nuclei, the cell lysate was gently pipetted over 2.5 volumes of chilled sucrose cushion (NP-40 lysis buffer in 24% w/v sucrose) and centrifuged for at 15 K $\times$ g for 10 minutes at 4°C. After removal of supernatant cytoplasmic extracts, the nuclei pellet was washed gently with 1 mL ice-cold 1X PBS/1 mM EDTA. The nuclei pellet was flicked to resuspend in 900 μ L of pre-chilled glycerol buffer (20 mM Tris-HCl [pH 8.0], 75 mM NaCl, 0.5 mM EDTA, 0.85 mM DTT, 1% protease and phosphatase inhibitor, 50% glycerol) before mixed with 3.6 mL of nuclei lysis buffer (10 mM Tris-HCl [pH 7.5], 1 mM DTT, 7.5 mM MgCl₂, 0.2 mM EDTA, 0.3 M NaCl, 1 M Urea, 1% NP-40, 1% protease and phosphatase inhibitor). After a 5 s vortex and a 10-min incubation on ice, the lysate was centrifuged at 15 K $\times$ g for 10 minutes at 4°C to pellet the chromatin. Finally, the chromatin pellet was rinse with PBS and then digested in 900 μ L chromatin extraction buffer (20 mM Tris-HCl [pH 7.5], 100 mM KCl, 2 mM MgCl₂, 2.5 mM CaCl₂, 0.3 M sucrose, and 0.1% v/v Triton X-100, 1% protease and phosphatase inhibitor) with 5 U/mL Micrococcal Nuclease (N3755, Sigma) by shaking at 4°C for 2 hours. 5 mM of EDTA was added to quench the reaction, and the chromatin lysate was cleared at 2 K $\times$ g for 10 minutes at 4°C.

For immunoprecipitation, the chromatin extract was combined with 5 μ g of rabbit anti-hnRNP antibody (Ab190352, abcam) and rotated overnight at 4°C. 100 μ L pre-washed Protein A beads (10002D, Thermo) was added to the reaction and rotated for another 4 hours at 4°C. The beads were then separated by being placed on a magnetic rack and washed with 1 mL IP wash buffer $\times$ 2 (50 mM Tris-HCl [pH 7.5], 300 mM NaCl, 0.1% v/v Triton X-100), 1 mL High-salt wash buffer $\times$ 2 (50 mM Tris-HCl [pH 7.5], 500 mM NaCl, 0.1% v/v Triton X-100), and 1 mL PNK buffer $\times$ 2 (50 mM Tris-HCl [pH 7.5], 50 mM NaCl, 10 mM MgCl₂). Co-immunoprecipitated RNA fragments were end-repaired on-beads by (1) 1 U/ μ L T4 PNK (EKO031, Thermo) in 100 μ L 1 $\times$ PNK buffer A (Thermo) at 37°C for 20 min with vigorous shaking; and then (2) 1 mM ATP (final concentration) with extra 0.5 U/ μ L T4 PNK at 37°C for another 20 min. After another two washes with PNK buffer, the beads were resuspended in 80 μ L 2 $\times$ Laemmli sample buffer and boiled at 95°C for 5 min.

The hnRNP-RNA complex was size-selected by SDS-PAGE (45-70 kDa). To extract the RNA, the gel slices were mashed and digested with 2 mg/ml protease K (25530049, Thermo) at 55°C for 1 hour. Then gel particles were filter out, and the RNA was purified by Acid-Phenol:Chloroform extraction and overnight ethanol precipitation. RNA libraries were generated using NEBNext multiplex small RNA library preparation kit (NEB, E7300S) and sequenced on NextSeq500 with single end 84-bp read length.

ChIP-seq

We followed the protocol of ChIP-IT[®] Express kit (53008, Active motif) with Spike-in normalization strategy. One 15 cm of HEK293T cells were collected 48 hours after transfection with 20 nM control siRNA (1027281, QIAGEN) or hnRNP siRNA (G1: SI00700077, QIAGEN; G2: SI00700084, QIAGEN). The cells were fixed on plates with 1% w/v formaldehyde (28906, Thermo) in the culture media for 10 min at room temperature with gently shaking. After quenching the crosslinking reaction by 125 mM glycine for 5 min, the cells were rinsed with ice-cold PBS twice, scraped by a cell lifter, and pelleted at 500 $\times$ g for 5 minutes at 4°C.

The chromatin was sheared according to the kit's manual. Briefly, the cell pellet was resuspended in 1 mL ice-cold "Lysis Buffer" and incubated on ice for 30 min. After homogenization with 30 strokes on ice in the dounce homogenizer, the nuclei were pelleted at 2.4 K $\times$ g for 10 minutes at 4°C. The nuclei pellet was then resuspended in 350 μ L "Shearing Buffer," made into aliquots of 100 μ L in 0.65 mL microtubes (C30010011, Diagenode), and sonicated with Bioruptor[®] Pico sonicator (20 s on, 30 s off, 10 cycles) to 200 to 500 bp in size. The sheared chromatin samples were cleared by centrifuge at 18 K $\times$ g for 10 minutes at 4°C. 10 μ L of the chromatin was heat-recrosslinked, treated with RNase A and Protease K to determine chromatin yield. The rest of the chromatin samples were adjusted to a concentration of 0.2 μ g/ μ L in the "Shearing Buffer."

The Spike-in chromatin (53083, Active motif) was added to final concentration of 0.1 ng/ μ L. 10 μ L of the chromatin with Spike-in was aliquoted and saved as "Input." For each of the siControl, sihnRNP-3, and sihnRNP-4 samples, a $\sim$ 200 μ L immunoprecipitation (IP) reaction was set up with the following components: 101 μ L chromatin with Spike-in, 25 μ L "Protein G Magnetic Beads," 20 μ L "ChIP Buffer 1," 2 μ L "Protease Inhibitor Cocktail," 1 μ L Spike-in antibody (61686, Active Motif), 46 μ L H₂O, and 5 μ g of the respective following antibodies: anti-Pol II (664912, BioLegend), anti-Pol II S2P (MAB10602, MBL life science), and anti-Pol II S5P (ab5131, abcam). The IP reactions were rotated overnight at 4°C before washed once with 800 μ L "ChIP Buffer 1" and twice with 800 μ L "ChIP Buffer 2." The "IP" chromatin was eluted, reverse cross-linked, and treated with RNase A and Protease K in parallel with "Input" according to the kit's manual. DNA was purified with DNA Clean & Concentrator-5 (D4013, Zymo research). DNA libraries

were generated with the KAPA Hyper Prep Kit (KK8502, KapaBiosystems) and sequenced on NextSeq500 with single end 84-bp read length.

RT-qPCR and RT-PCR

Total RNA, collected for mRNA sequencing as described above, was reverse transcribed using the SuperScript III First-Strand Synthesis System (18080-051, Thermo). For validation of changes in transcript abundance, reverse transcription was followed by quantitative PCR using Power SYBR Green PCR Master Mix (4367659, Thermo Scientific) under the following conditions: 95°C for 10 minutes; 40 cycles of [95°C for 15 s, 60°C for 1 minute]. For validation of splicing changes, the cDNA was amplified using Taq DNA Polymerase (EP0401, Thermo) under the following conditions: 95°C for 3 minutes, 30 cycles of [95°C for 30 s, 50°C for 30 s, 72°C for 1 minute], and 72°C for 10 minutes. The RT-PCR products were resolved on a 10% polyacrylamide Novex TBE gel (EC62762BOX, Thermo) and stained with SYBR Gold Nucleic Acid Gel Stain (S11494, Thermo).

QUANTIFICATION AND STATISTICAL ANALYSIS

Western blot quantification

For quantification of co-immunoprecipitation, western blot bands were quantified using ImageLab software, and RNAPII bands were normalized to hnRNPG bands and to input as follows: $(\text{RNAPII}_{\text{IP}} / \text{hnRNPG}_{\text{IP}}) / (\text{RNAPII}_{\text{input}} / \text{hnRNPG}_{\text{input}})$, where X_Y is the intensity of band X in lane Y .

Gel quantification

For limited proteolysis, bands on denaturing protein gels were quantified using ImageJ, and the fractions of full-length hnRNPG in each lane were calculated and fit to a single exponential function to derive the half-life ($t_{1/2}$).

For RT-PCR validation of splicing, bands on native acrylamide gels were quantified using ImageJ and used to calculate the ratio of exon inclusion to exon exclusion.

mRNA sequencing analysis

The mRNA sequencing data from this study (NCV, WT, RRMmut, RGG1mut, RGG2mut) and from two previous studies were aligned to the human genome version hg19 using the STAR computer program version 2.5.3a (Dobin et al., 2013) with soft-clipping, yielding approximately 266 million (this study, Table S1) and 132 million (Liu et al., 2017) mapped reads per sample. Gene and exon boundaries from RefSeq (O'Leary et al., 2016) were extracted from the University of California Santa Cruz (UCSC) table browser (Karolchik et al., 2004) for human genome version hg19. Differential expression levels of exons and genes in RefSeq were analyzed using featureCounts (Liao et al., 2014) followed by DESeq2 (Love et al., 2014). The fold change (FC) in gene and exon expression levels was calculated as $\log_2(\text{Counts}_{[\text{NCV or mut}]} / \text{Counts}_{\text{WT}})$ for NCV, RRMmut, RGG1mut, and RGG2mut, or as $\log_2(\text{Counts}_{\text{KD}} / \text{Counts}_{\text{Control}})$ for hnRNPG, METTL3, and METTL14 knockdowns (KD). Biological relevance (fold change) and statistical significance were considered simultaneously via the π value = $-\log_2(\text{FC}) \cdot \log_{10}(p\text{-value})$ (Xiao et al., 2014). Differentially expressed genes and exons were selected using the threshold $|\pi \text{ value}| \geq 0.4292$, which corresponds to significance level $\alpha = 0.1$ under assumption of independence between fold change and p -value (volcano plots show broad range of p -values for a given fold change, and a broad range in fold change for a given p -value). Co-down- or co-upregulated exons were exons that were differentially expressed, with $\log_2(\text{FC}) < 0$ or $\log_2(\text{FC}) > 0$, respectively, both upon hnRNPG KD and upon either METTL3 KD or METTL14 KD. Differentially spliced exons were identified as differentially expressed exons (DEE) for which $\log_2(\text{FC}_{\text{DEE}}) - \log_2(\text{FC}_{\text{nonDEE}}) > 0.9 \cdot \log_2(\text{FC}_{\text{DEE}})$ for some non-differentially expressed exon (nonDEE) in the same gene. Based on this analysis, the majority of differentially expressed exons were also differentially spliced exons, and differentially spliced exons showed the same patterns of down- versus upregulation as differentially expressed exons: RRMmut sequencing yielded 3191 downregulated and 894 upregulated differentially spliced exons, RGG1mut yielded 503 down- and 433 upregulated differentially spliced exons, and RGG2mut yielded 3574 down- and 944 upregulated differentially spliced exons relative to WT. For correlation plots comparing changes in gene or exon expression in different samples, genes or exons that were differentially expressed in either NCV, RRMmut, RGG1mut, or RGG2mut based on the $|\pi \text{ value}| \geq 0.4292$ threshold were combined. The union of differentially expressed genes or exons was plotted to compare the $\log_2(\text{FC})$ in two different sequencing samples relative to WT. The Pearson correlation coefficient r , and the associated p -value based on the Fisher transformation, were calculated using R statistical software. Model II major axis linear regression was performed using lmodel2 in R statistical software. Gene ontology (GO) analysis was performed using the enrichment analysis tool provided by the Gene Ontology Consortium (Ashburner et al., 2000; Mi et al., 2017; The Gene Ontology Consortium, 2017).

Analysis of hnRNPG-bound m⁶A site distribution

Reads from previously published photoactivatable ribonucleoside-enhanced crosslinking and immunoprecipitation (PAR-CLIP) followed by methylated RNA immunoprecipitation (MeRIP) sequencing data (Liu et al., 2017) were aligned to the human genome version hg19 using the STAR computer program version 2.5.3a (Dobin et al., 2013) with soft-clipping, yielding approximately 17.7 million reads (standard deviation: 15.2 million) for each sample. The hnRNPG-bound m⁶A sites were identified based on two criteria. First, hnRNPG binding sites were identified based on PAR-CLIP T-to-C mutation profiles uncovered using PARalyzer version 1.1

(Corcoran et al., 2011). Second, hnRNPG-bound m⁶A sites were identified within hnRNPG binding sites based on positive enrichment of m⁶A-immunoprecipitated fragment counts relative to input at RRACH sites. Positive enrichment was defined using the threshold π value = $-\log_2(\text{FC}) \cdot \log_{10}(\text{p-value}) \geq 0.4292$, where $\log_2(\text{FC})$ is the winscore defined by (Dominissini et al., 2012) at RRACH sites found in hg19 reference chromosomal sequences (direct search for RRACH sequence motif for sites on + strand, and direct search for DGTYY motif for sites on – strand). For the analysis of the distribution of hnRNPG-bound m⁶A sites around the splice sites of differentially regulated exons, we identified all the hnRNPG-bound m⁶A sites with position x relative to the splice site (defined as position 0) such that $-300 \leq x \leq +300$ nucleotides, for either the 3' or 5' splice sites of any of the differentially regulated exons in the selected category. For instance, for the red curve in Figure 4D, we identified all hnRNPG-bound m⁶A sites that occurred within 300 nucleotides of the 3' splice site of any of the exons co-downregulated upon hnRNPG KD and either METTL3 KD or METTL14 KD. For each category of differentially expressed exons, the distribution of hnRNPG-bound m⁶A sites around each splice site was converted by kernel density estimation using R statistical software. The resulting density at each position was multiplied by the constant n/m , where n is the total number of hnRNPG-bound m⁶A sites within 300 nucleotides of the splice site, and m is the total number of differentially expressed exons in the selected category. The resulting value at each position corresponds to the mean number of hnRNPG-bound m⁶A sites per exon at that position, among all differentially expressed exons in the selected category. These values (sites/exon) were plotted at each position x , with $-250 \leq x \leq +250$, in Figures 3C, 4E, and S6C. Rapid identification of overlapping genomic features was made possible by BEDTools intersect function version 2.17.0 (Quinlan and Hall, 2010).

ChIP-seq data Analysis

Pol II, Pol II S2P and Pol II S5P ChIP-seq reads were sent to trim-galore to remove adaptor contamination and low-quality sequences, trimmed sequences were mapped with Bowtie (v2.2.6) using default parameters. Duplicate reads were removed by samtools rmdup -s. Unique and monoclonal mapped reads were extended to 150 bp based on the average sonicated chromatin DNA length. Spike-in reads were mapped to *Drosophila* genome(dm6) using same parameters. Pol II, Pol II S2P and Pol II S5P peaks were called using MACS2 with broad peak mode and default settings($q\text{value} = 0.01$). genomeCoverageBed (bedtools) was used to transfer bed files to bedGraph files, then Spike-in read numbers were used as a normalization factor that equalizes the signal across samples. We used deeptools to do the Pol II, Pol II S2P and Pol II S5P distribution analysis on 5' and 3' splice sites regions (–250bp - 250bp). (ComputeMatrix reference-point-referencePoint center -R hnRNPG_regulated_exon.5/3splicesite.bed -S siC-Pol2.bigWig sihnG1-Pol2.bigWig sihnG2-Pol2.bigWig -b 250 -a 250-skipZeros-binSize 1-samplesLabel siC sihnG1 sihnG2-missingDataAsZero -p 30 -o Pol2_on_hnRNPG_regulated_exon.5/3splicesite.matrix.mat.gz & plotProfile -m Pol2_on_hnRNPG_regulated_exon.5/3splicesite.matrix.mat.gz-plotHeight 7-plotWidth 9-samplesLabel siC sihnG1 sihnG2-colors red blue green-perGroup -out Pol2_on_hnRNPG_regulated_exon.5/3splicesite_Profile.png &).

DATA AND CODE AVAILABILITY

The sequencing data have been deposited to the National Center for Biotechnology Information Gene Expression Omnibus database under accession number GEO: GSE114311. Mendeley data are available at <https://doi.org/10.17632/5hrrjww3xc.1>. Additional processed data are available from the authors upon request.